

Processamento de Imagens

Transformada de Fourier (Parte II)

Professora Leo Sampaio Ferraz Ribeiro



Slide para não esquecer de passar a lista

[illegible]

Ar Condicionado está ok?



Dúvidas de Trabalhos e Exercícios?



Transformada Discreta de Fourier 2D

$$F(u, v) = \frac{1}{mn} \sum_{x=0}^{n-1} \sum_{y=0}^{m-1} f(x, y) e^{-j2\pi \left(\frac{ux}{n} + \frac{vy}{m} \right)}$$

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O ângulo da phase é definido por:

$$\phi(u, v) = \arctan \left(\frac{I(u, v)}{R(u, v)} \right)$$

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Diretamente da equação temos o “componente DC”:

$$F(0,0) = \frac{1}{nm} \sum_{x=0}^{n-1} \sum_{y=0}^{m-1} f(x, y)$$

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O espectro fourier de uma função real é simétrico em relação a origem:

$$|F(u, v)| = |F(-u, -v)|$$

E o ângulo da fase é anti-simétrico

$$\phi(u, v) = -\phi(-u, -v)$$

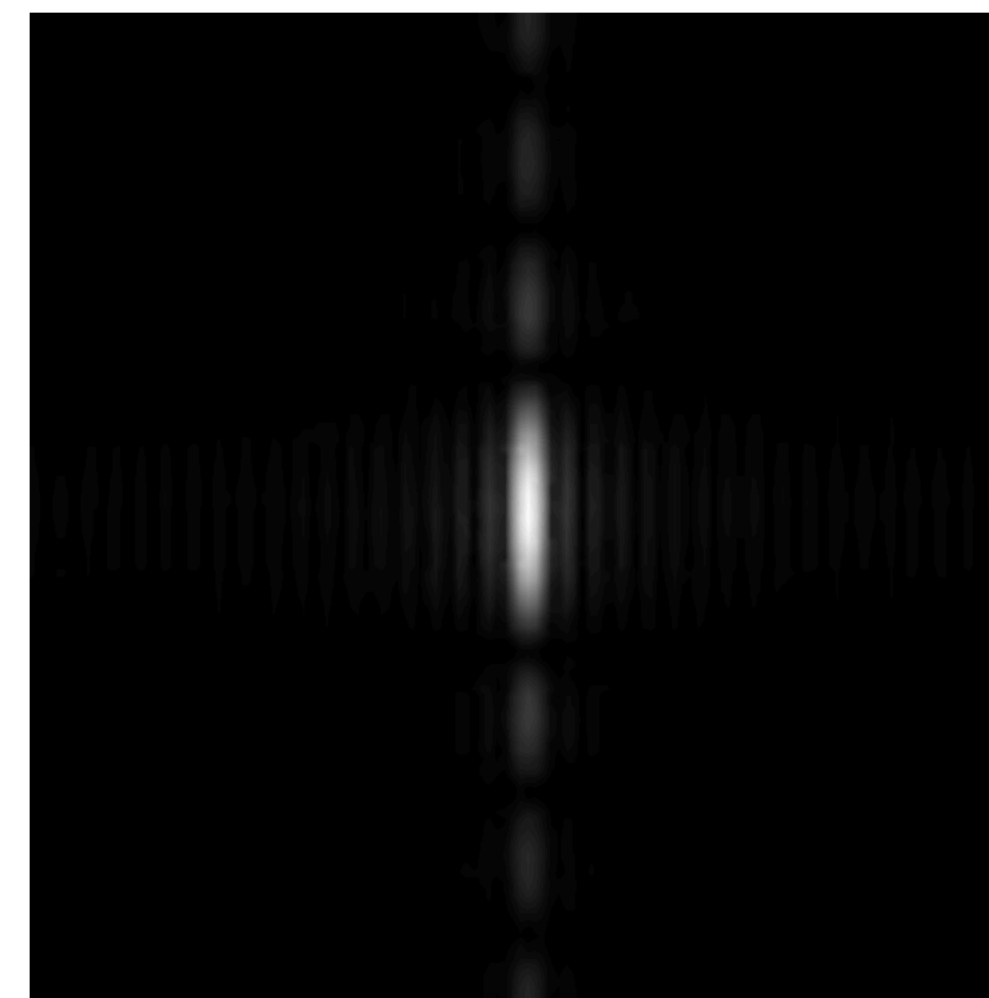
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Para mostrar uma função transformada é comum reposicionar os quadrantes usando coordenadas $(-1)^{x+y}$ e aplicar log:



$|F(u, v)|$



$\text{Shift}(|F(u, v)|)$

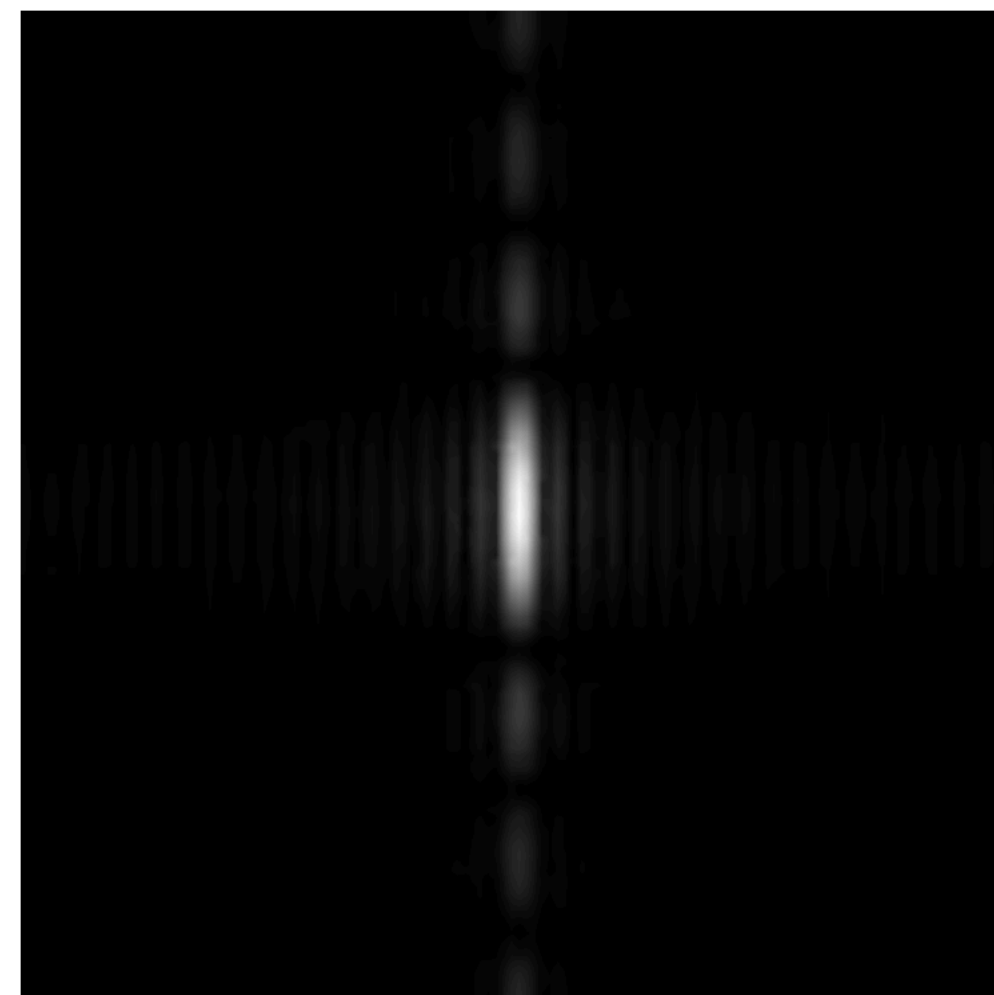
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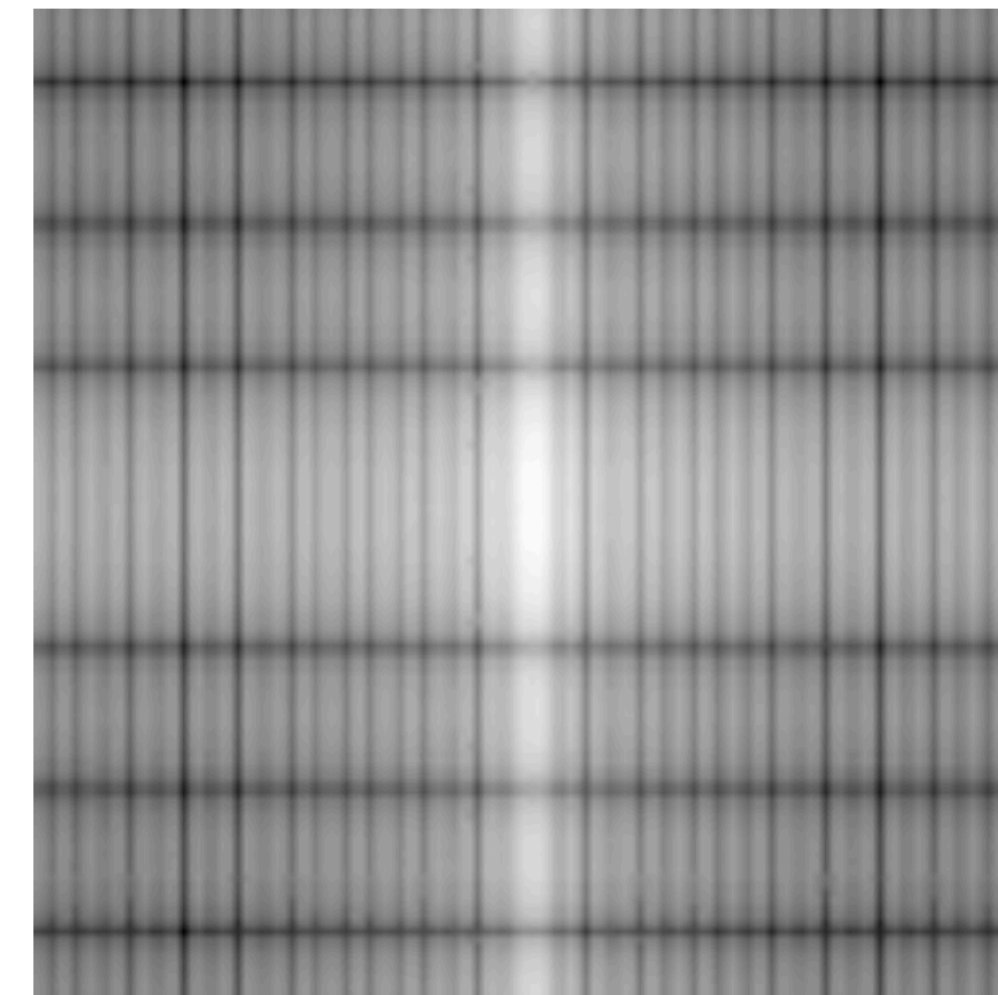
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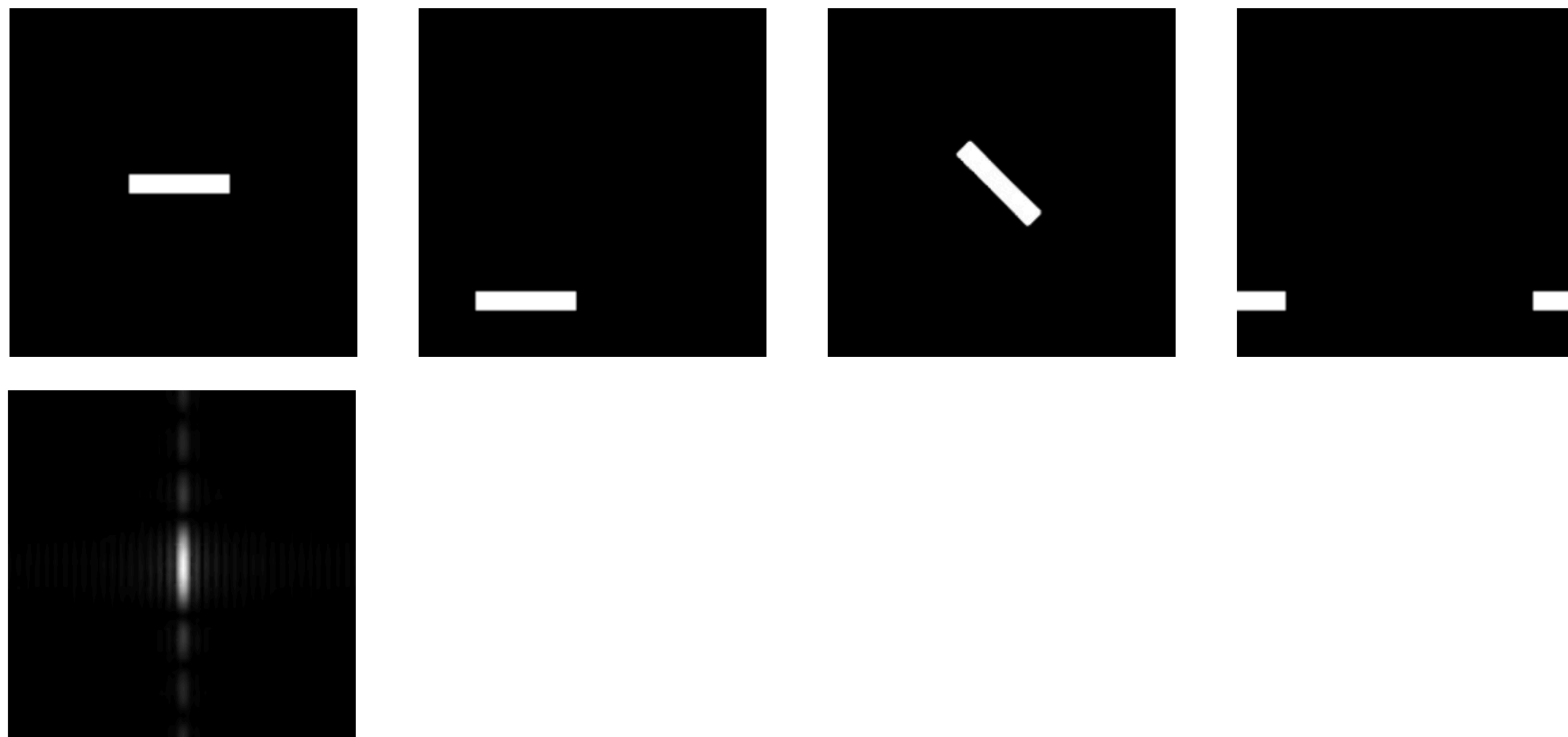
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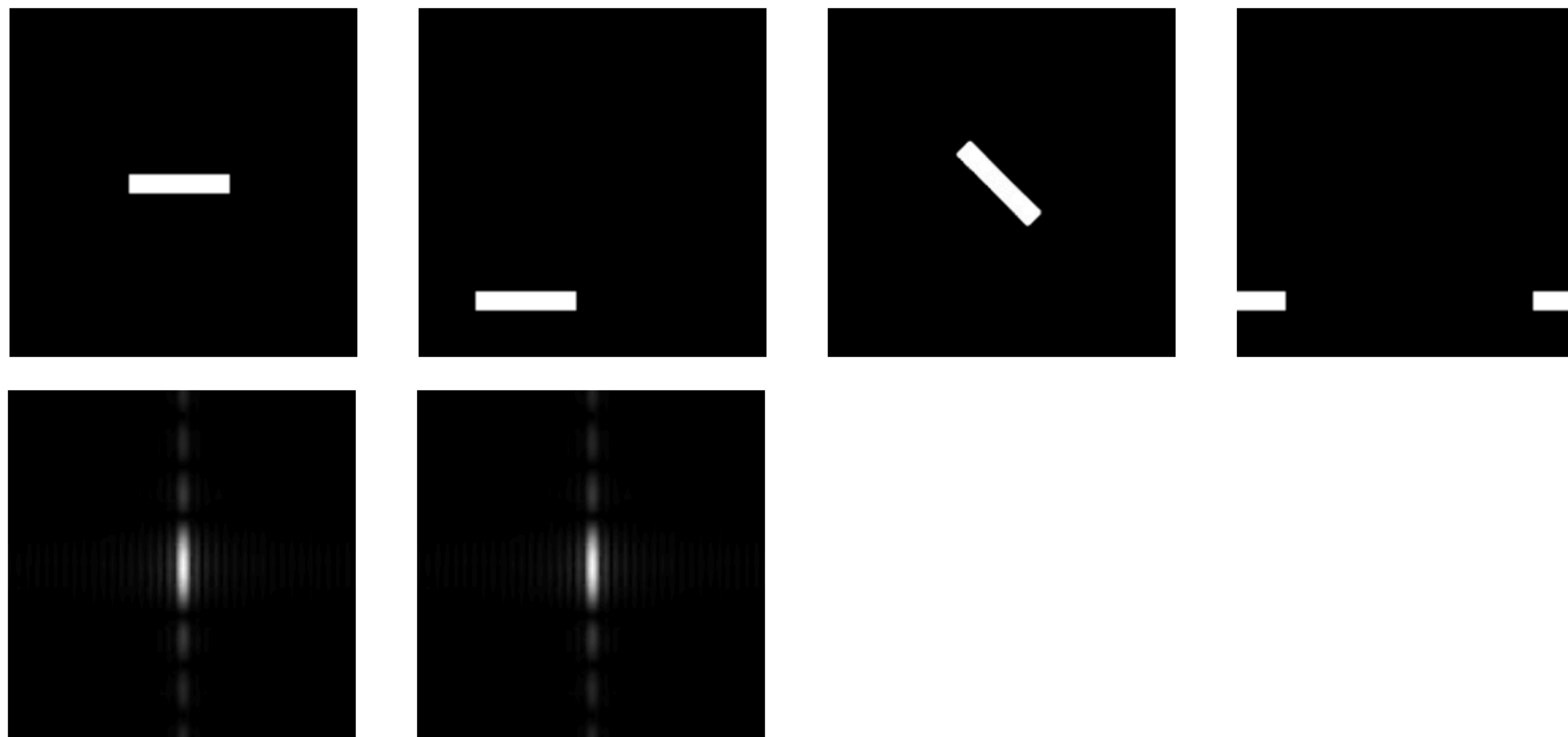
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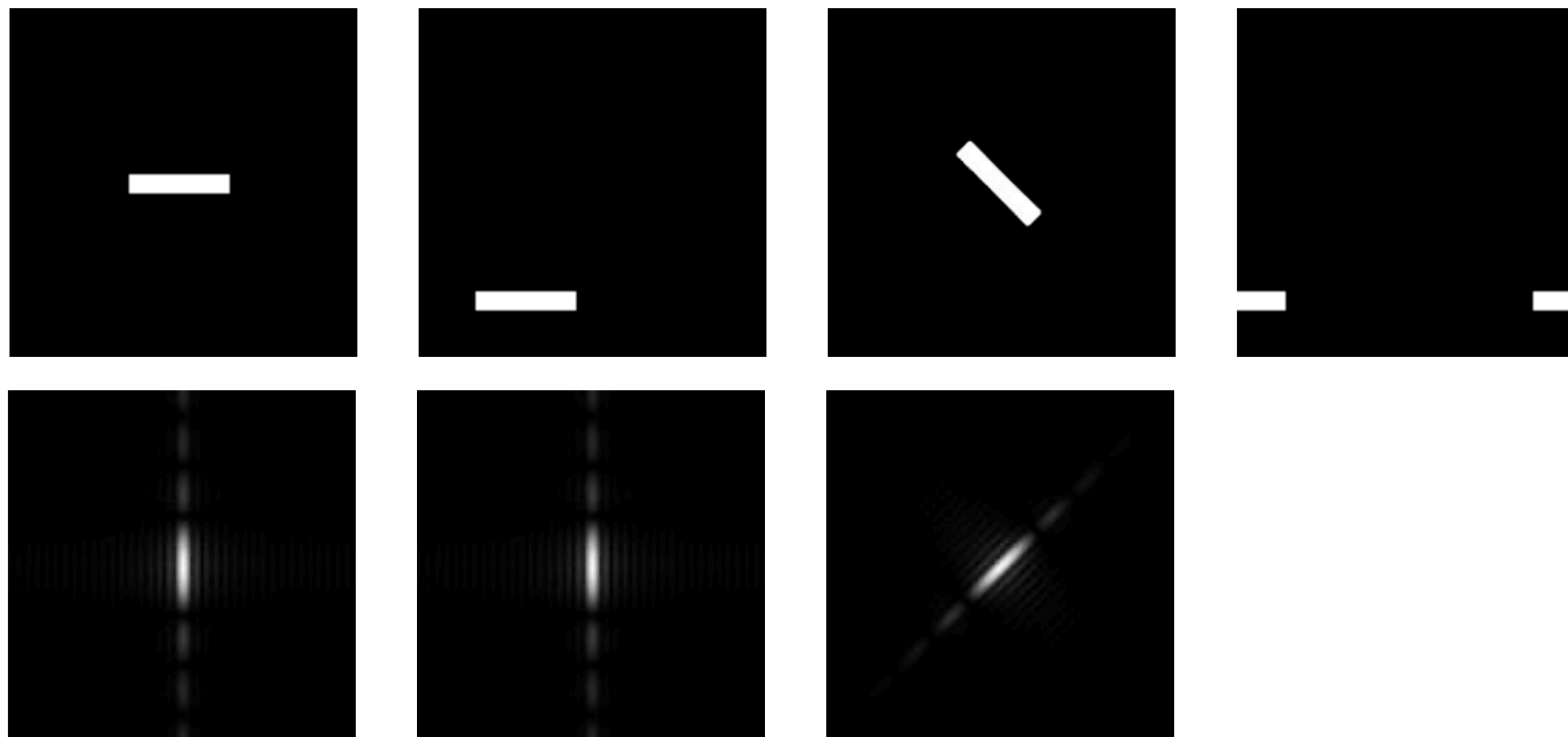
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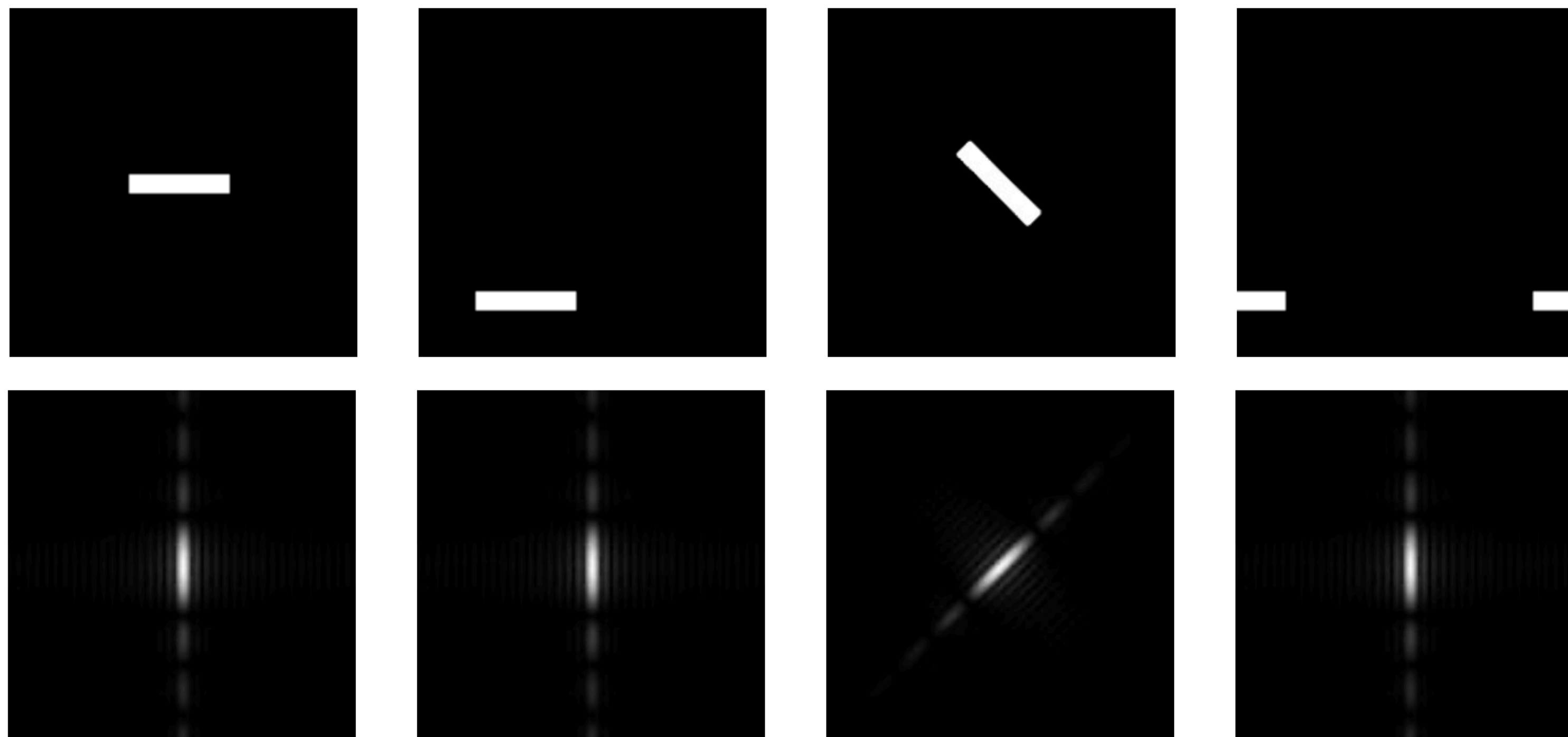
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Transformada Discreta de Fourier 2D

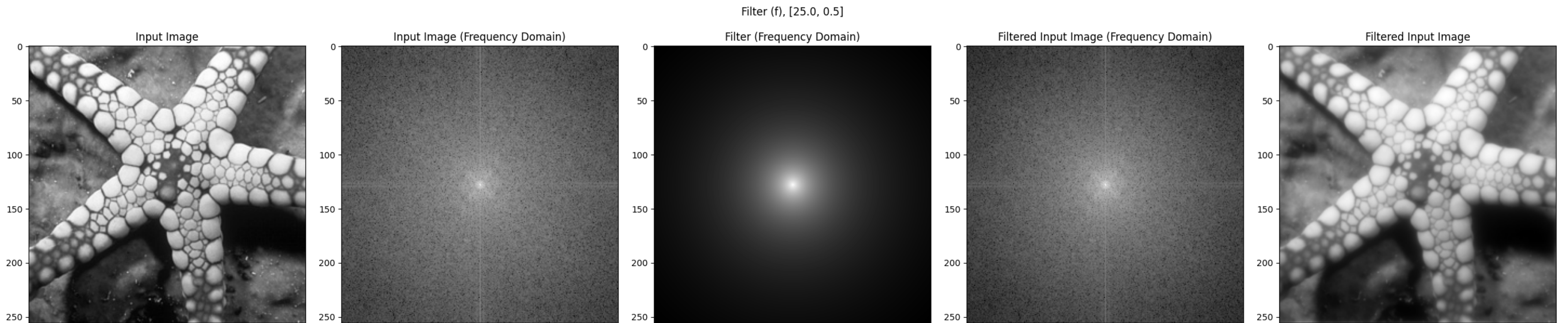
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Após o shift nós temos no centro da imagem as frequências mais baixas



Teorema da Convolução

$$(f \circledast h)(x, y) = \sum_{i=0}^{n-1} \sum_{j=0}^{m-1} f(i, j) h((x - i), (y - j))$$

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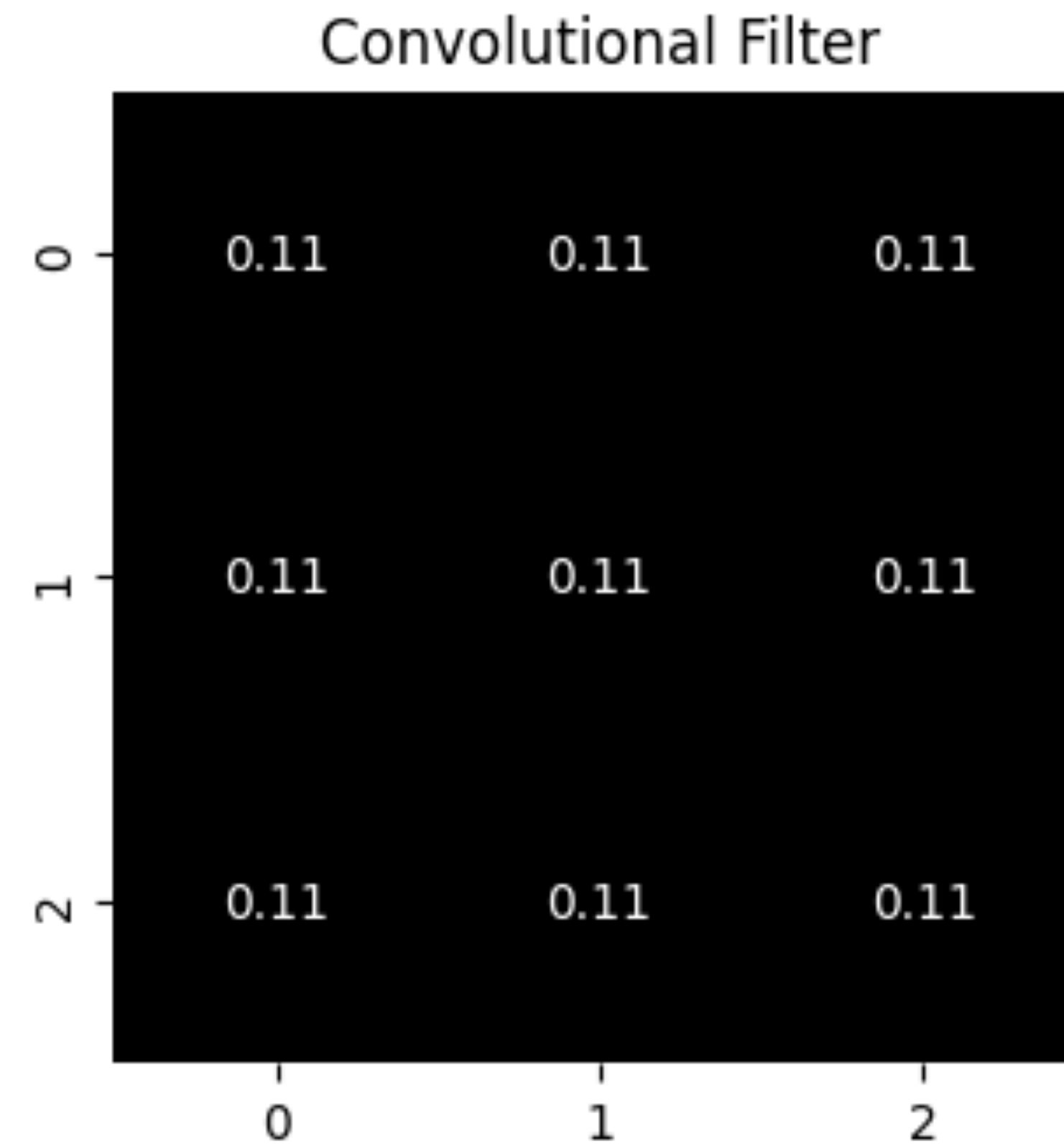
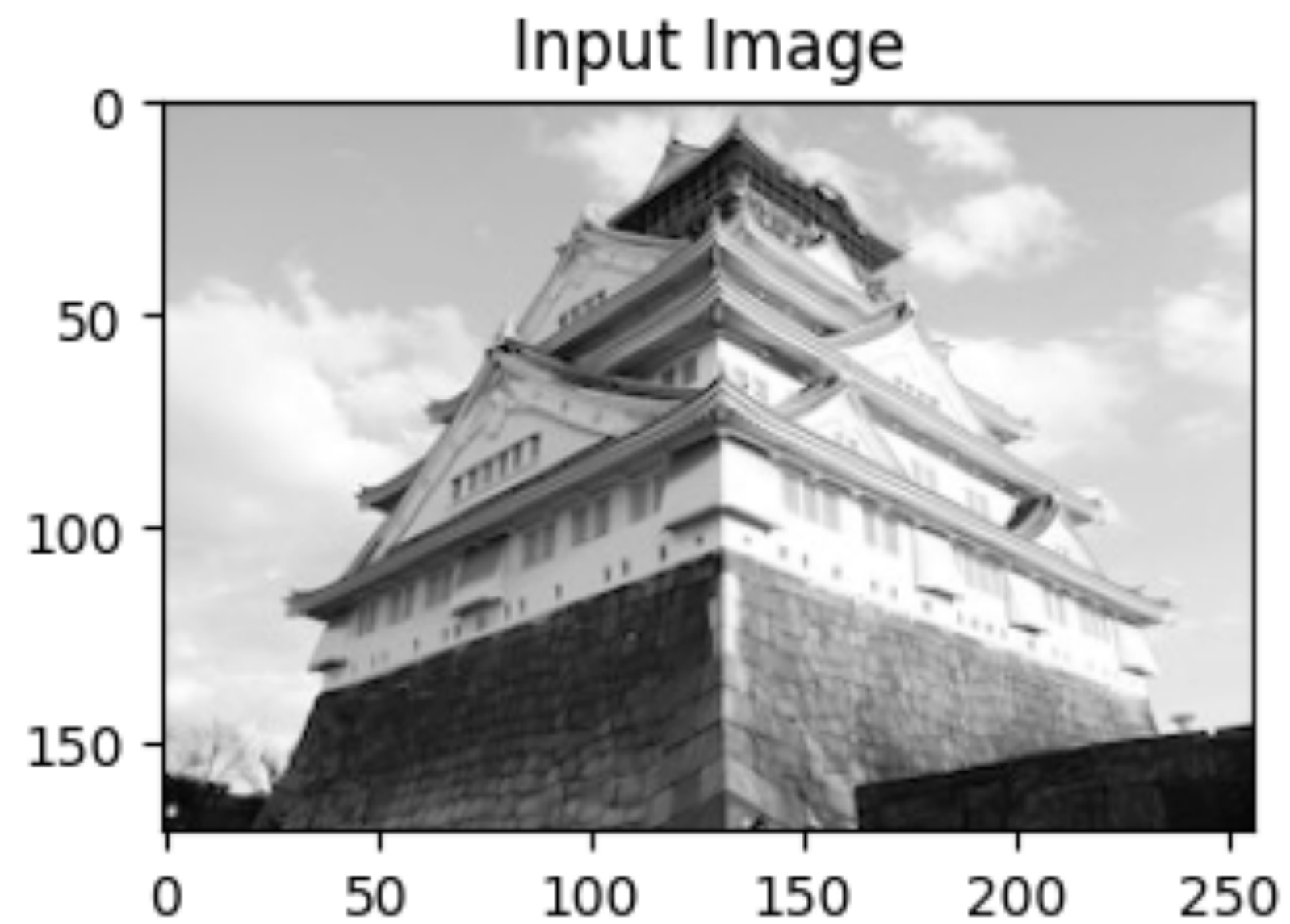
Uma convolução no domínio espacial/temporal é equivalente a uma multiplicação no domínio das frequências

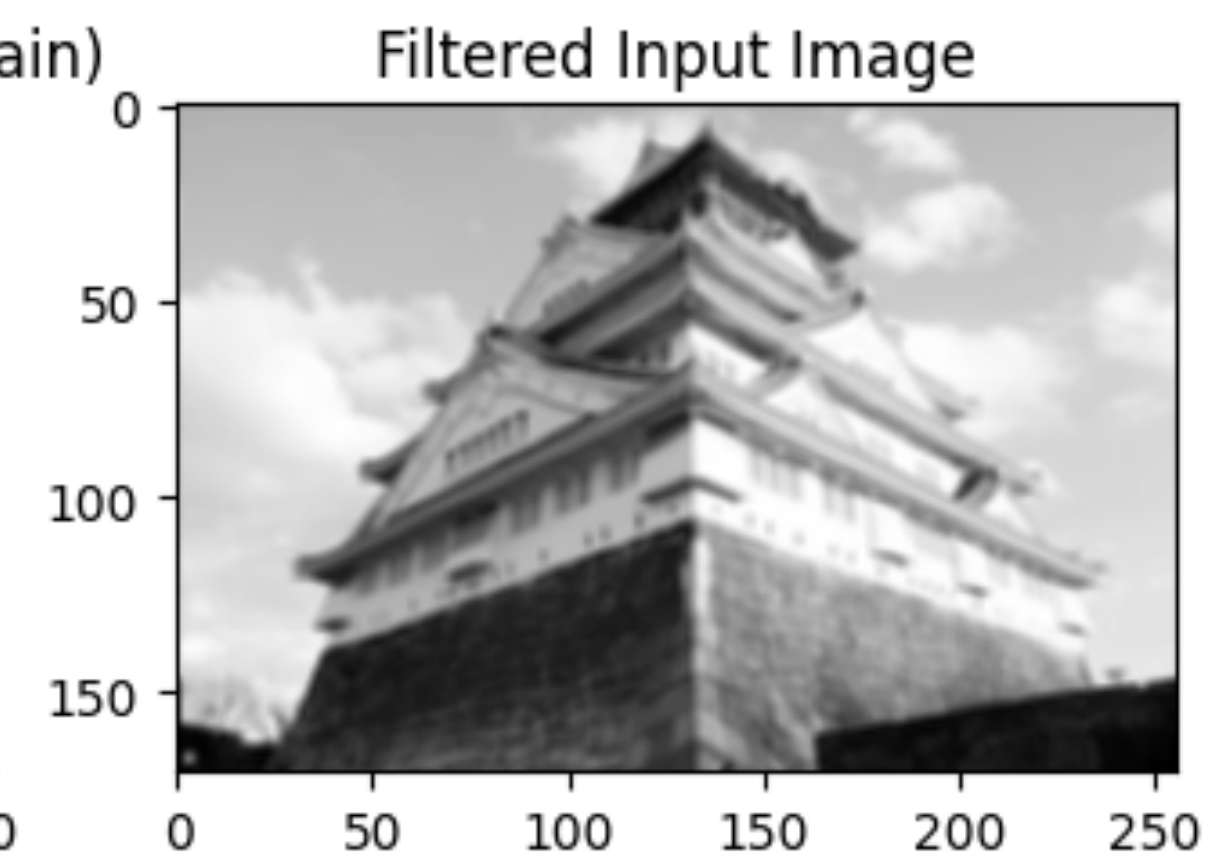
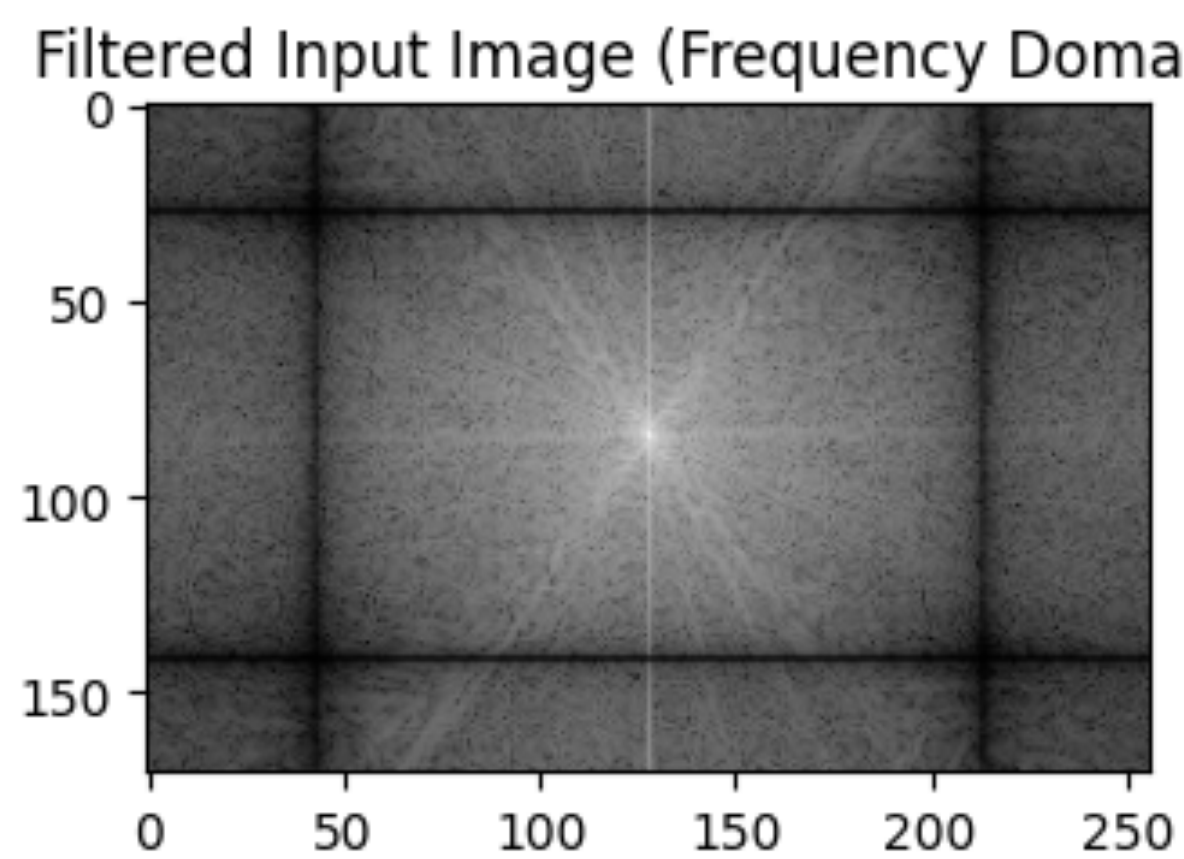
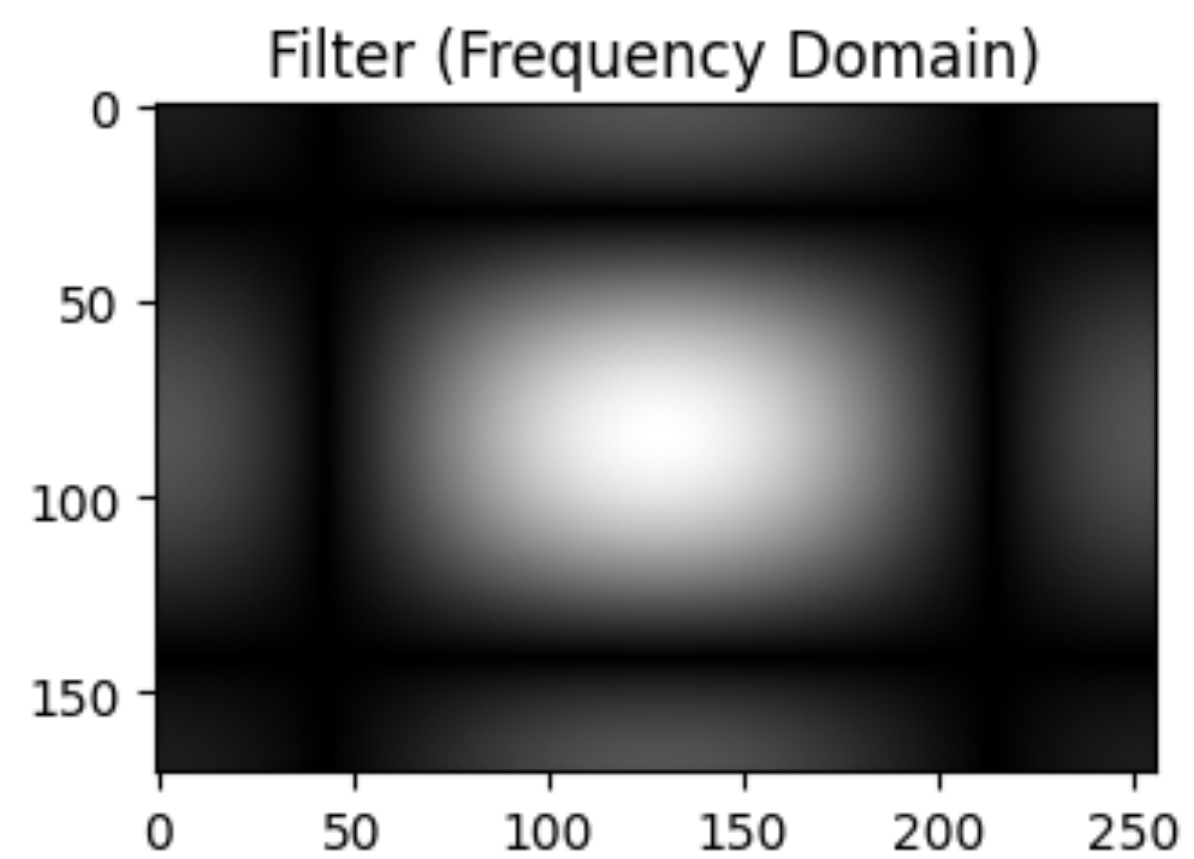
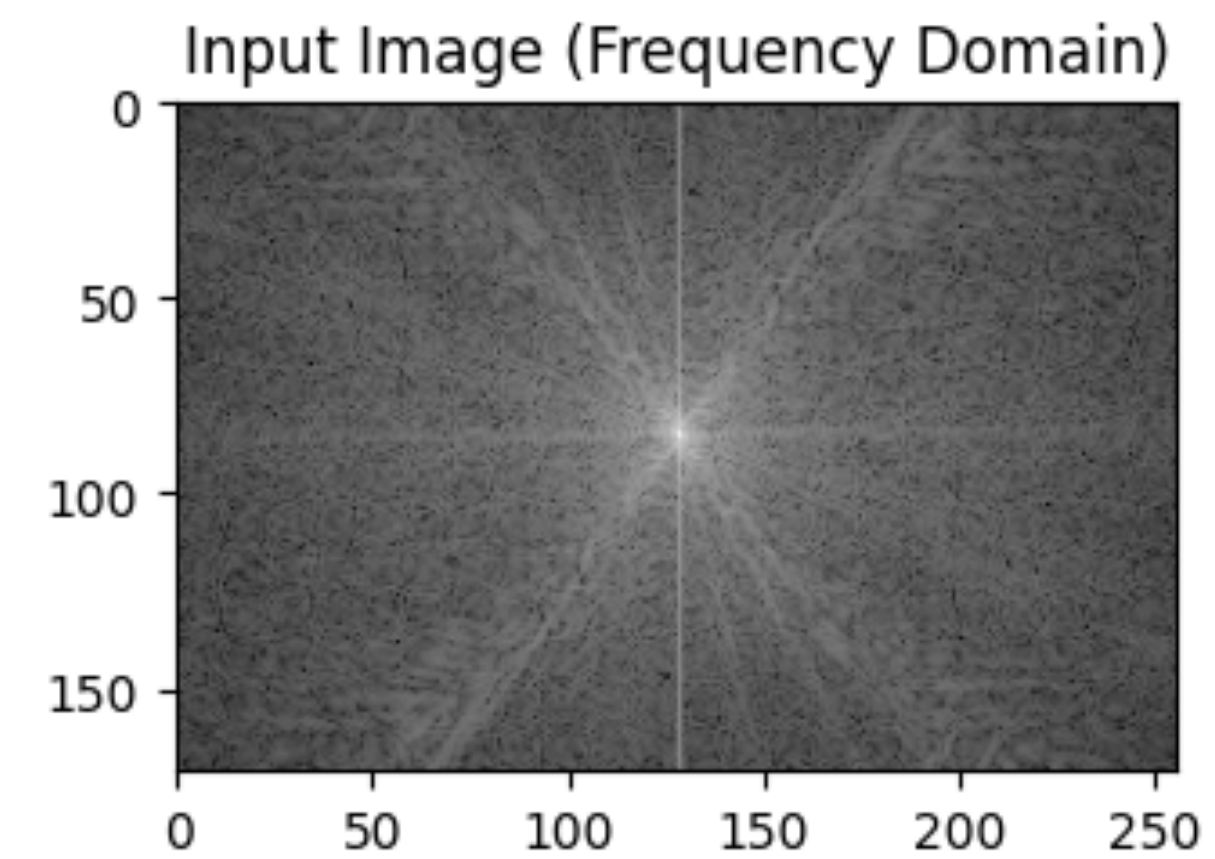
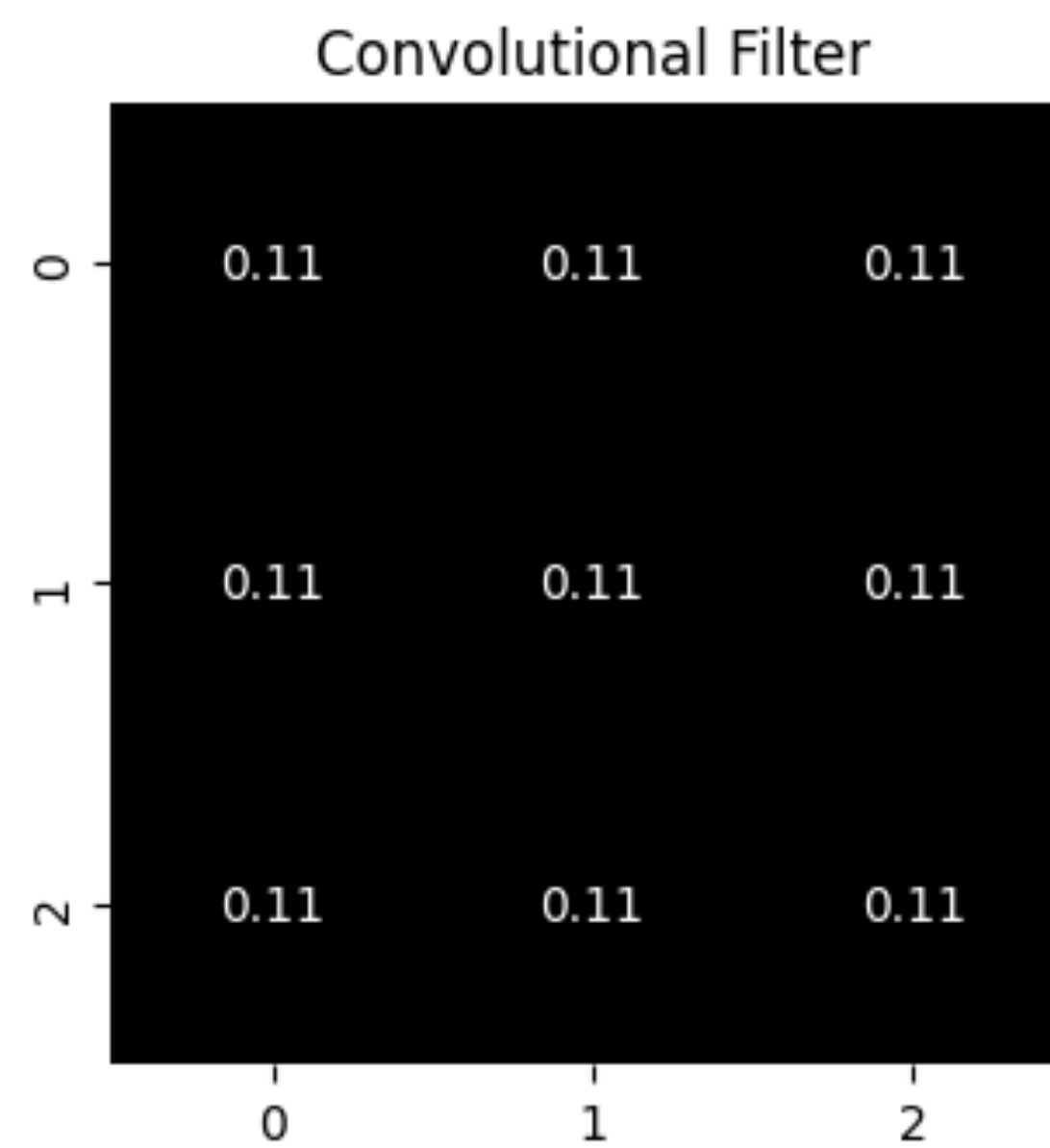
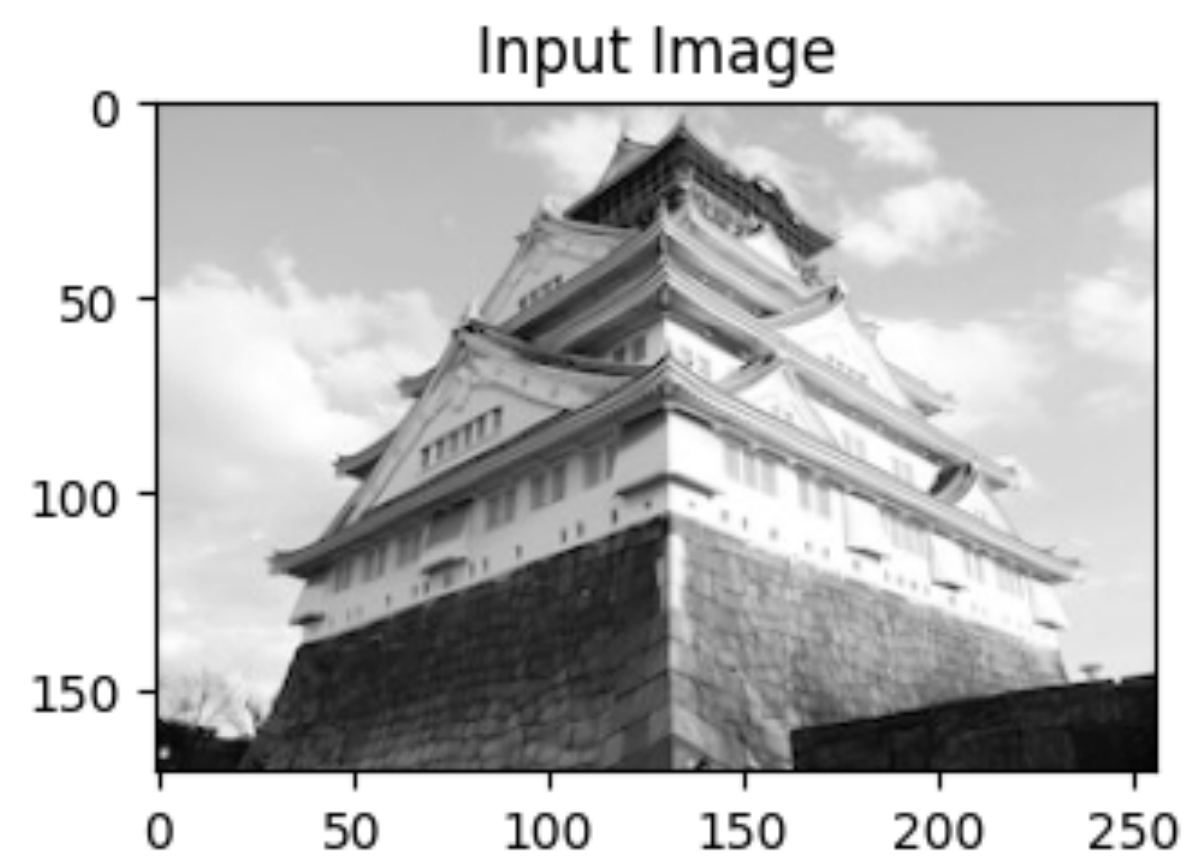
$$f(x, y) \circledast h(x, y) \leftrightarrow F(u, v) \cdot H(u, v)$$

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Filtros no Domínio das Frequências

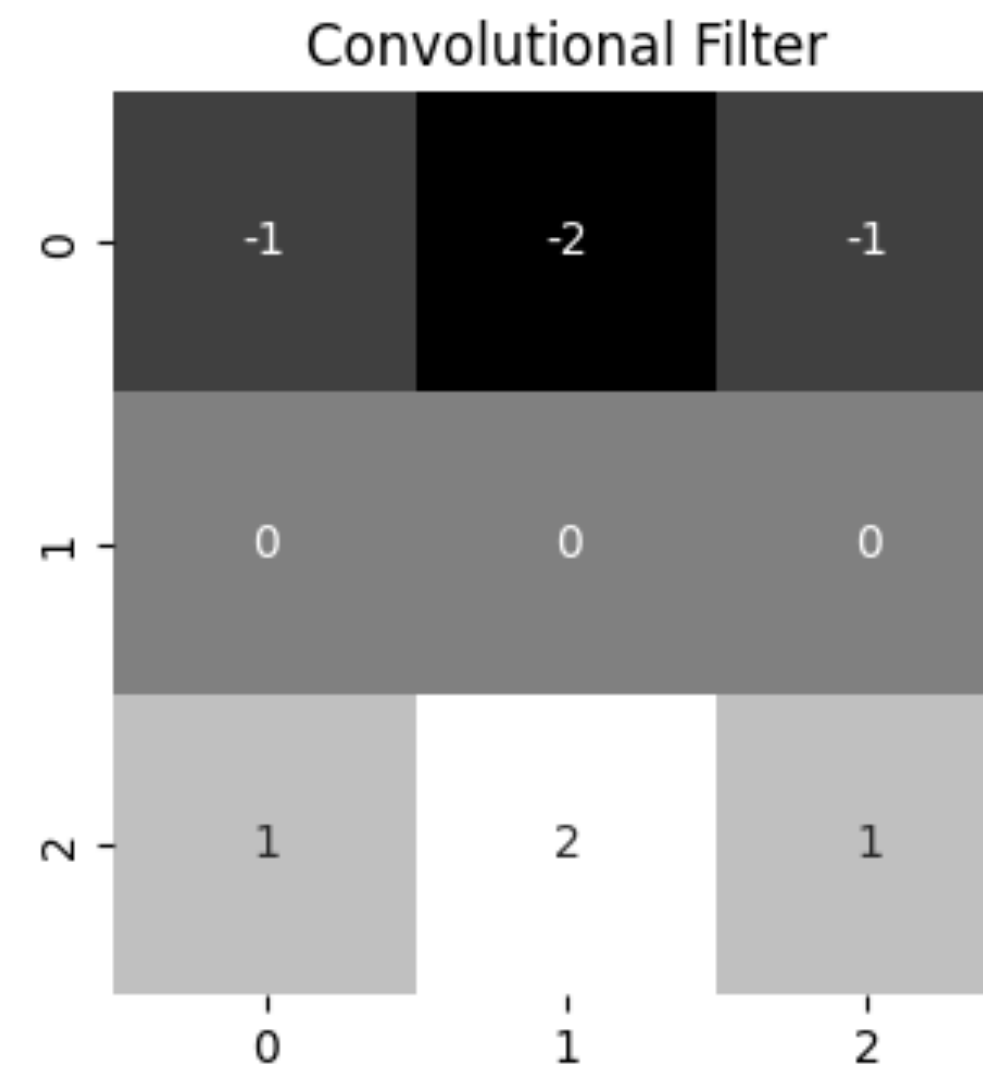
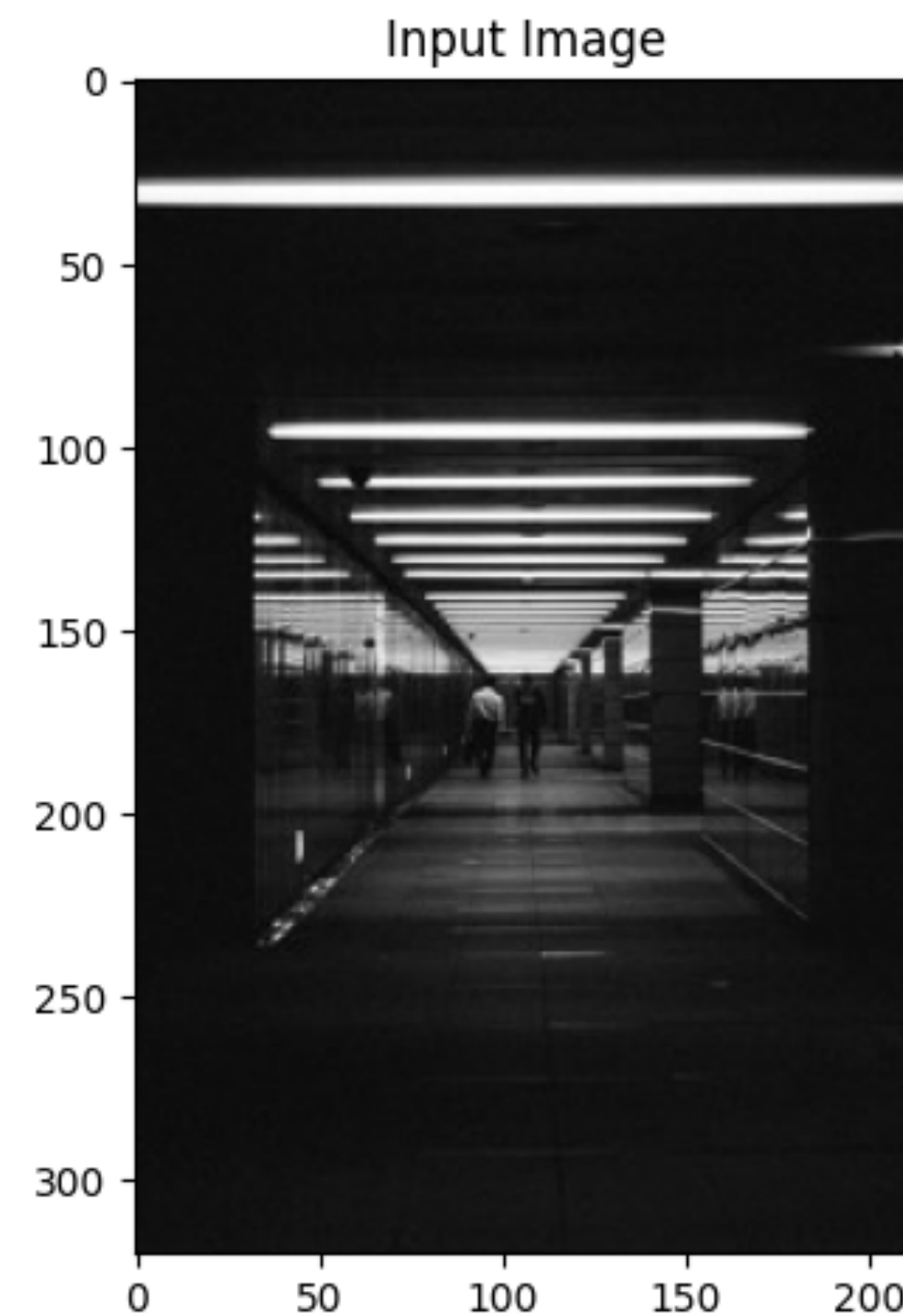
Filtro caixa

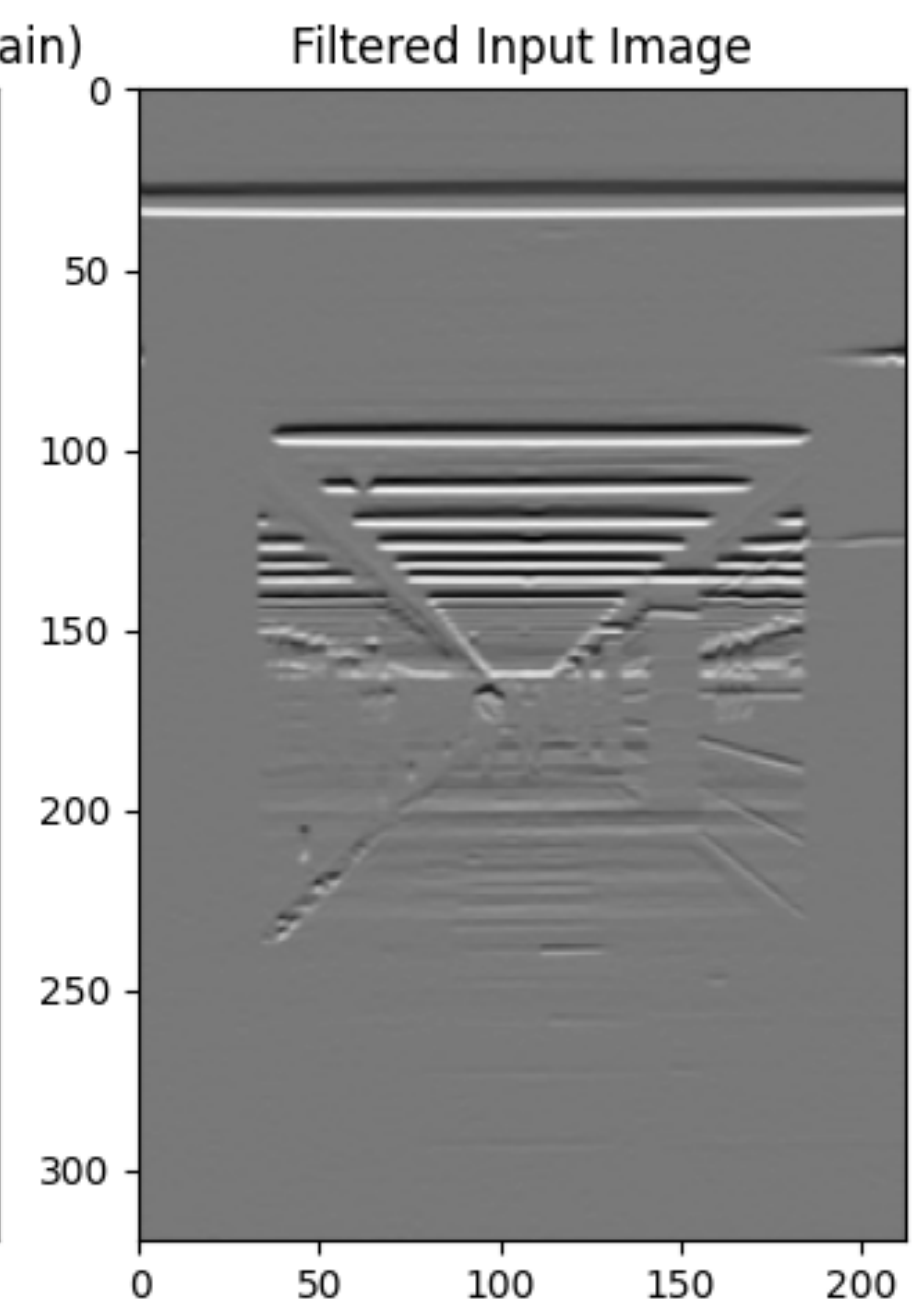
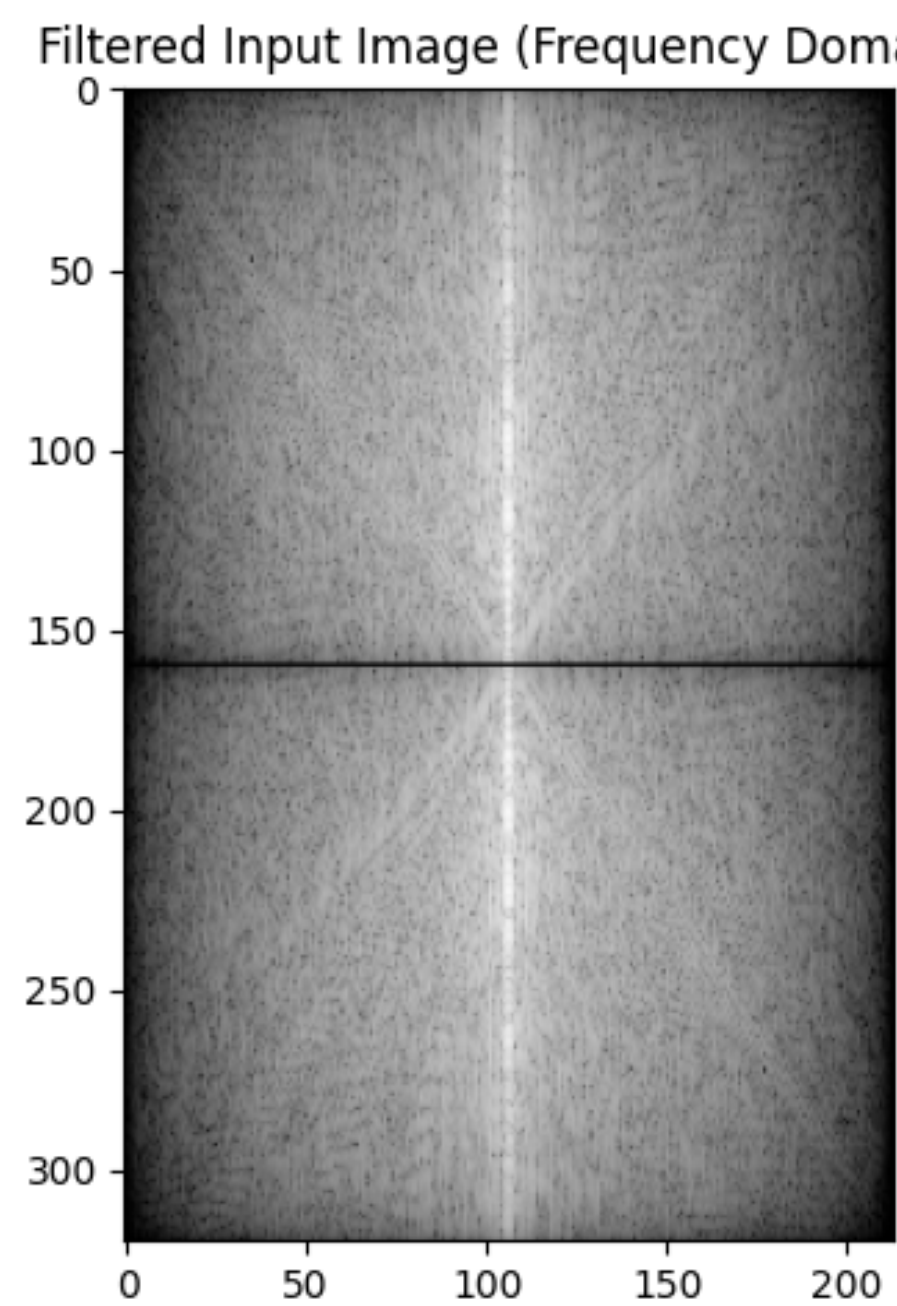
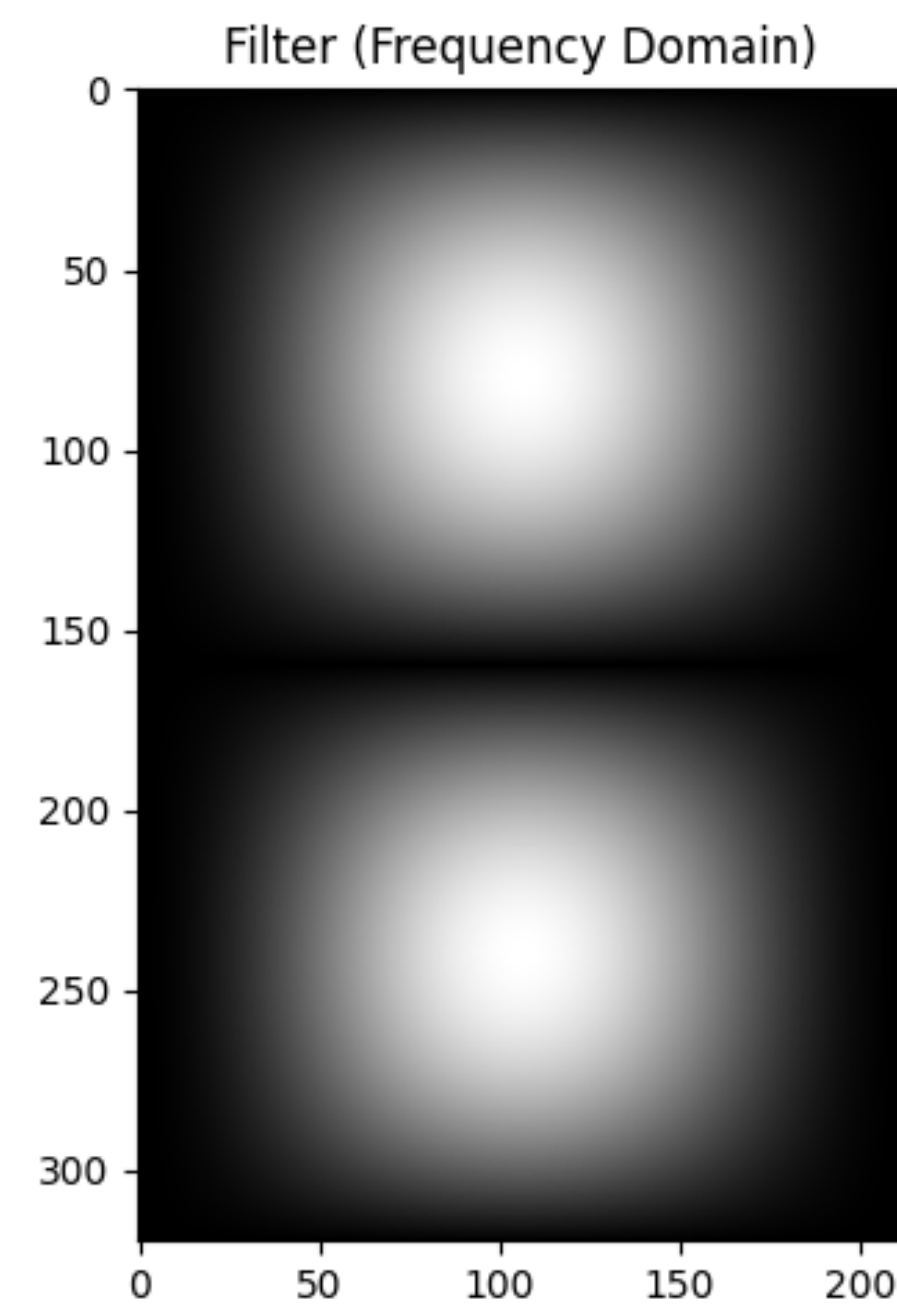
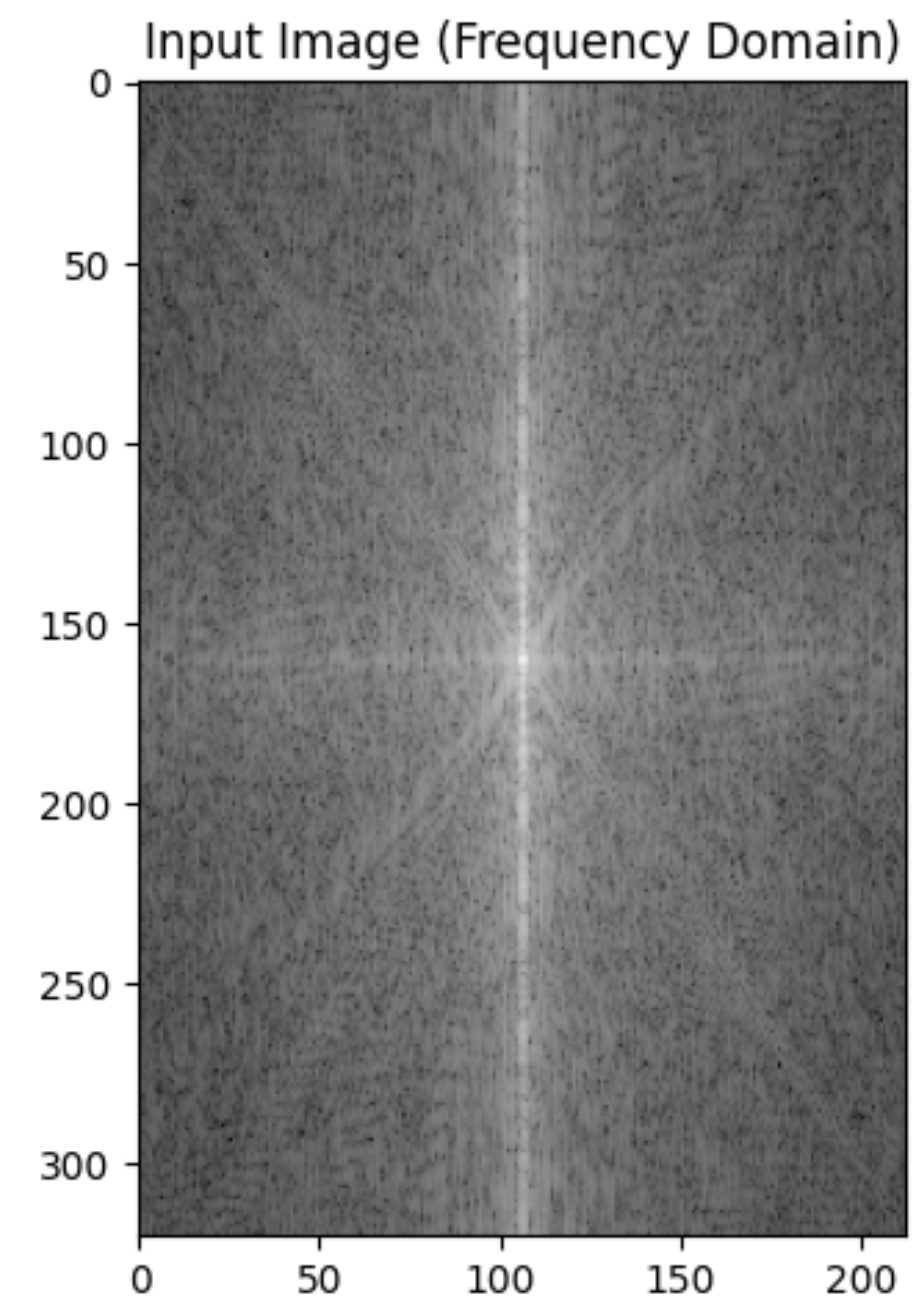
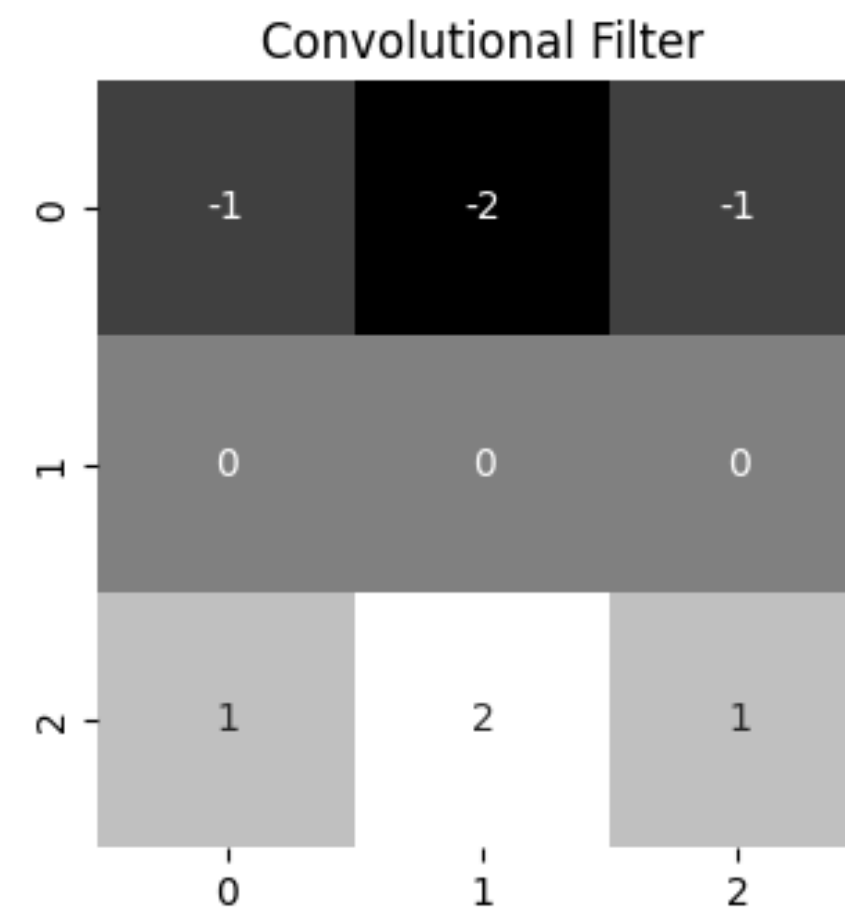
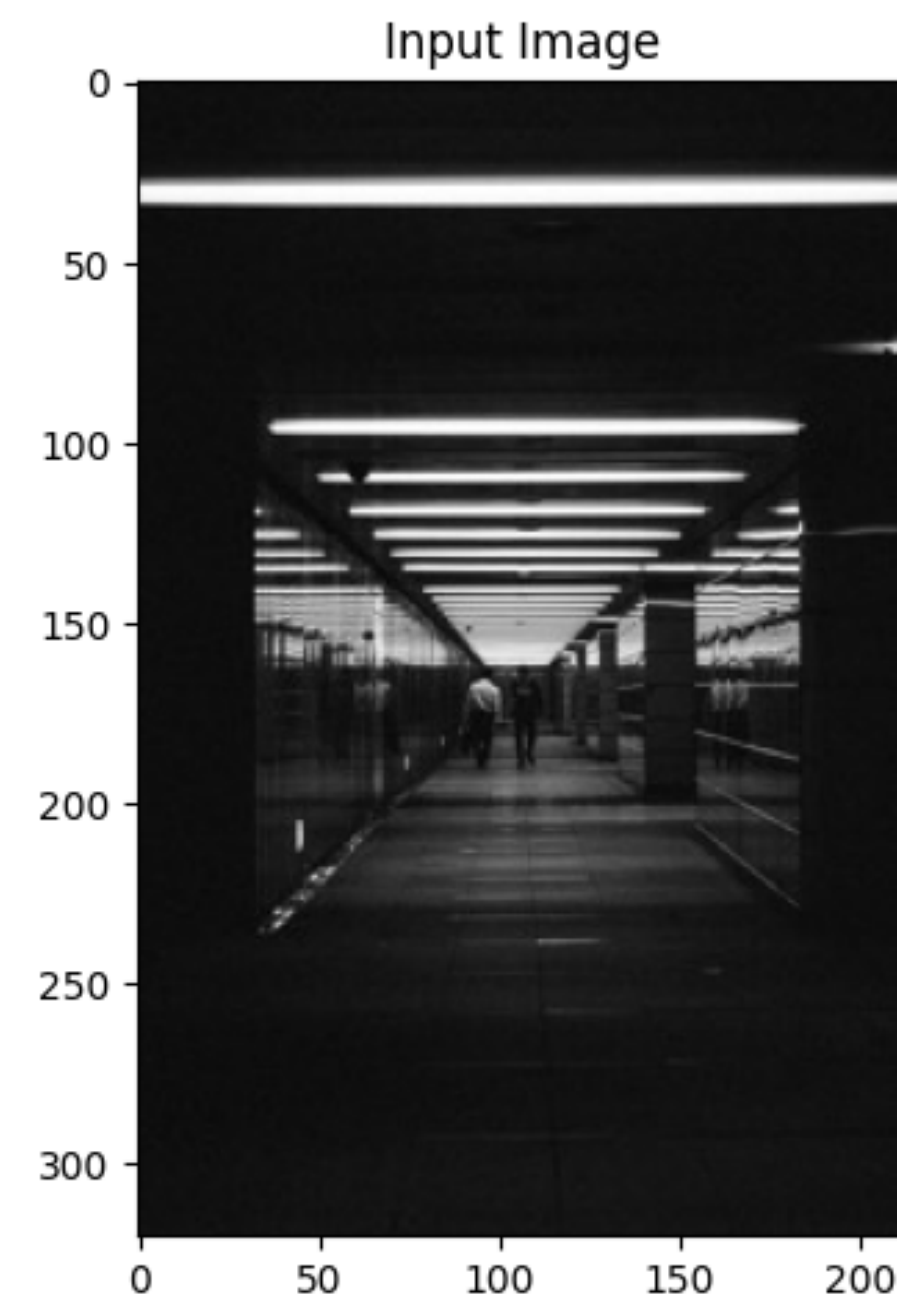




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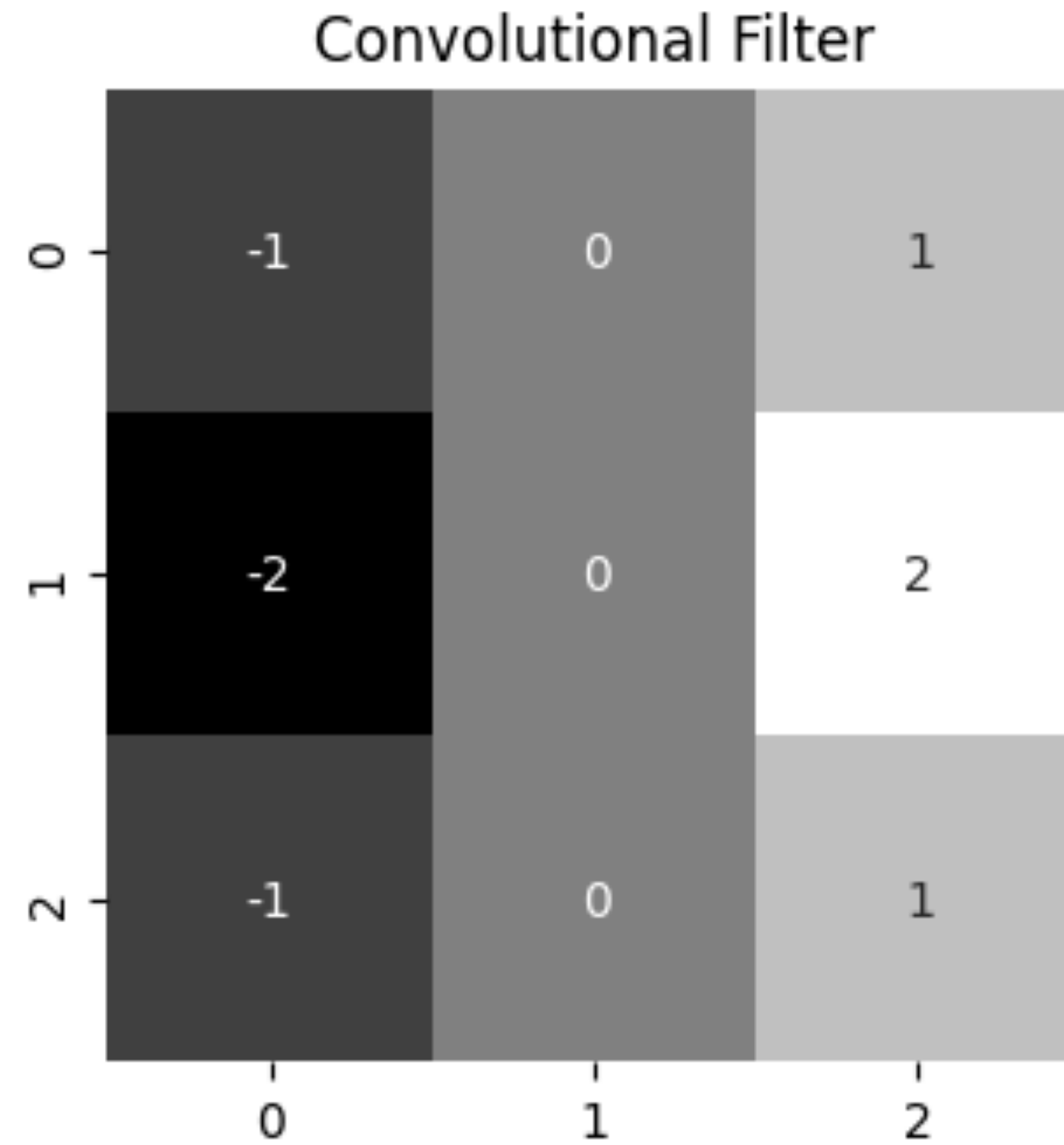
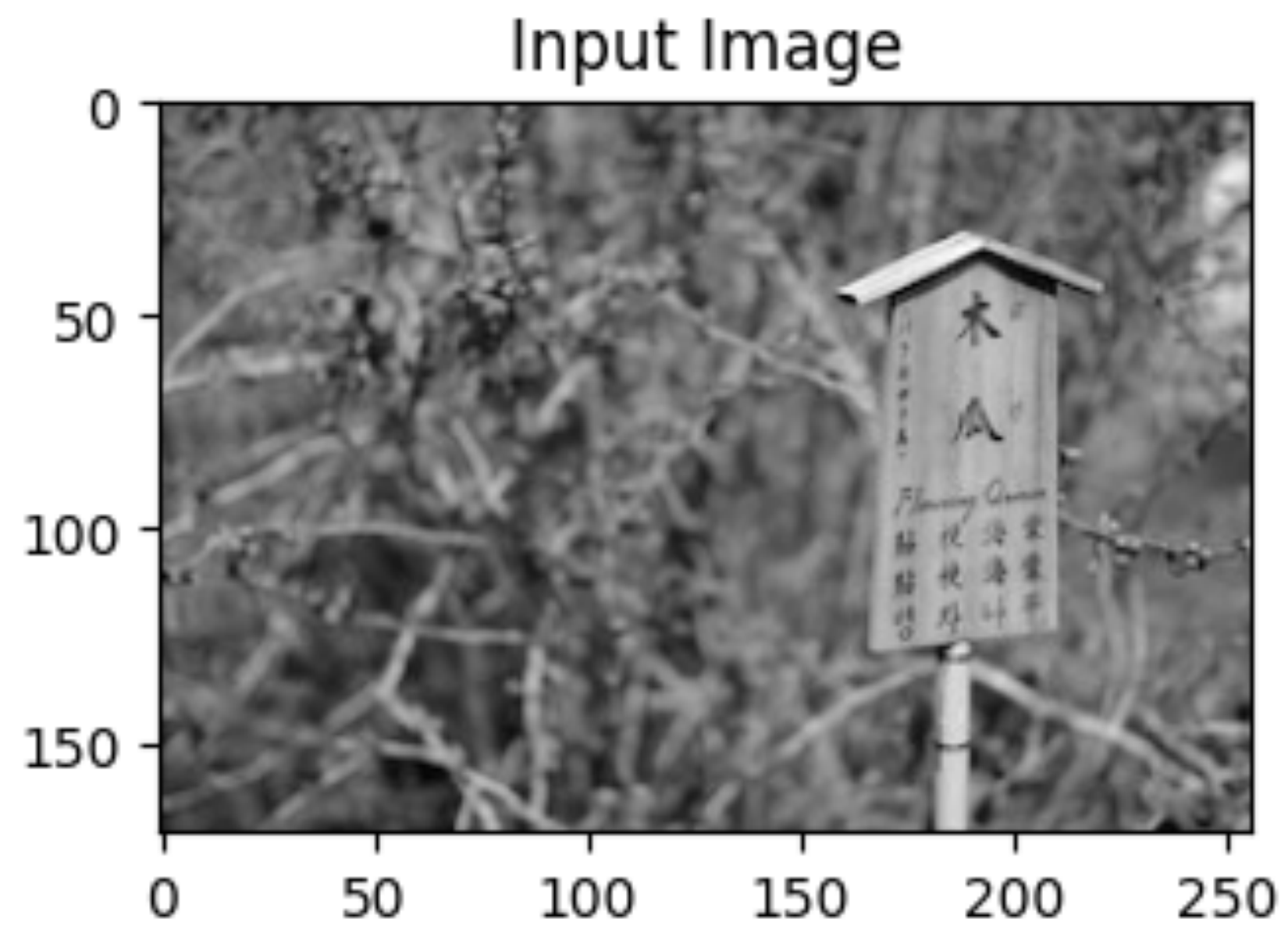
Filtro Sobel

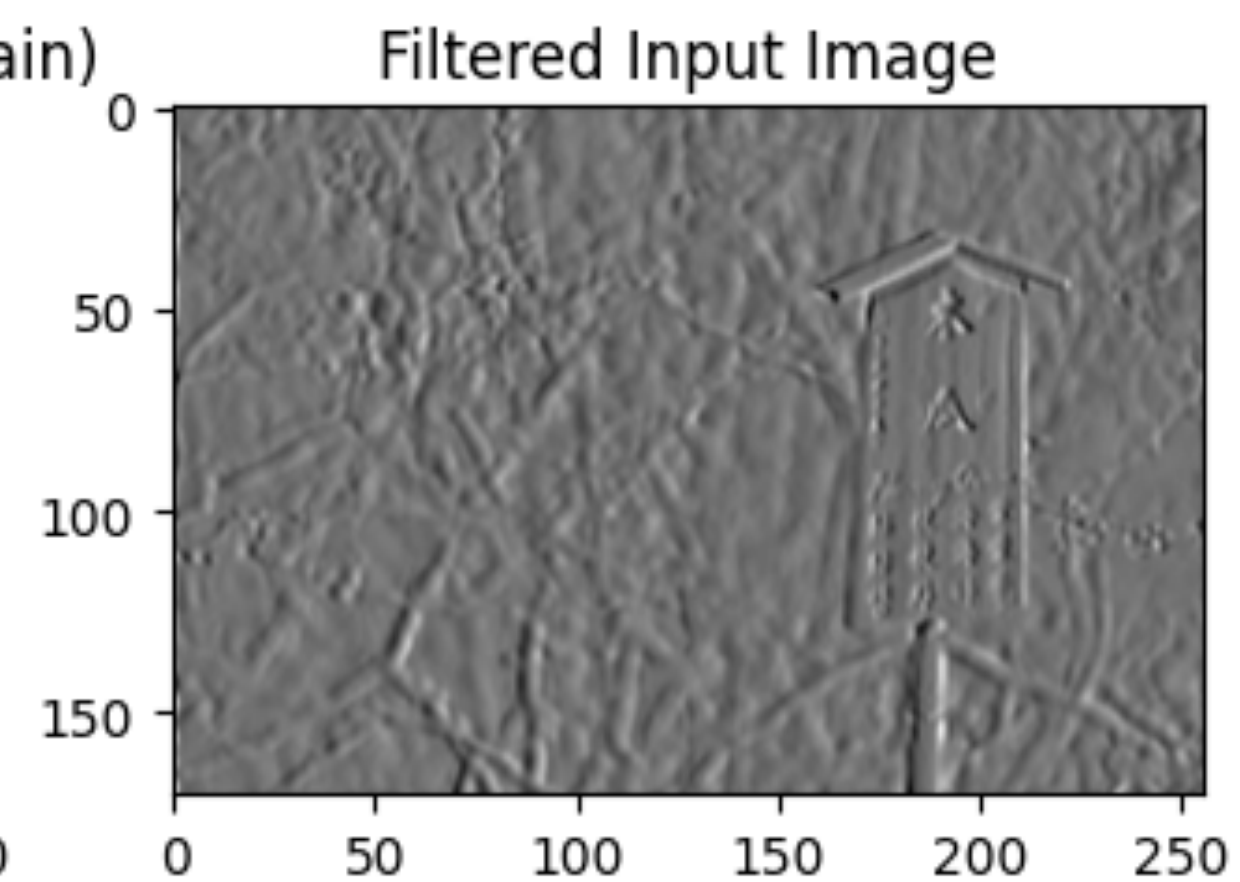
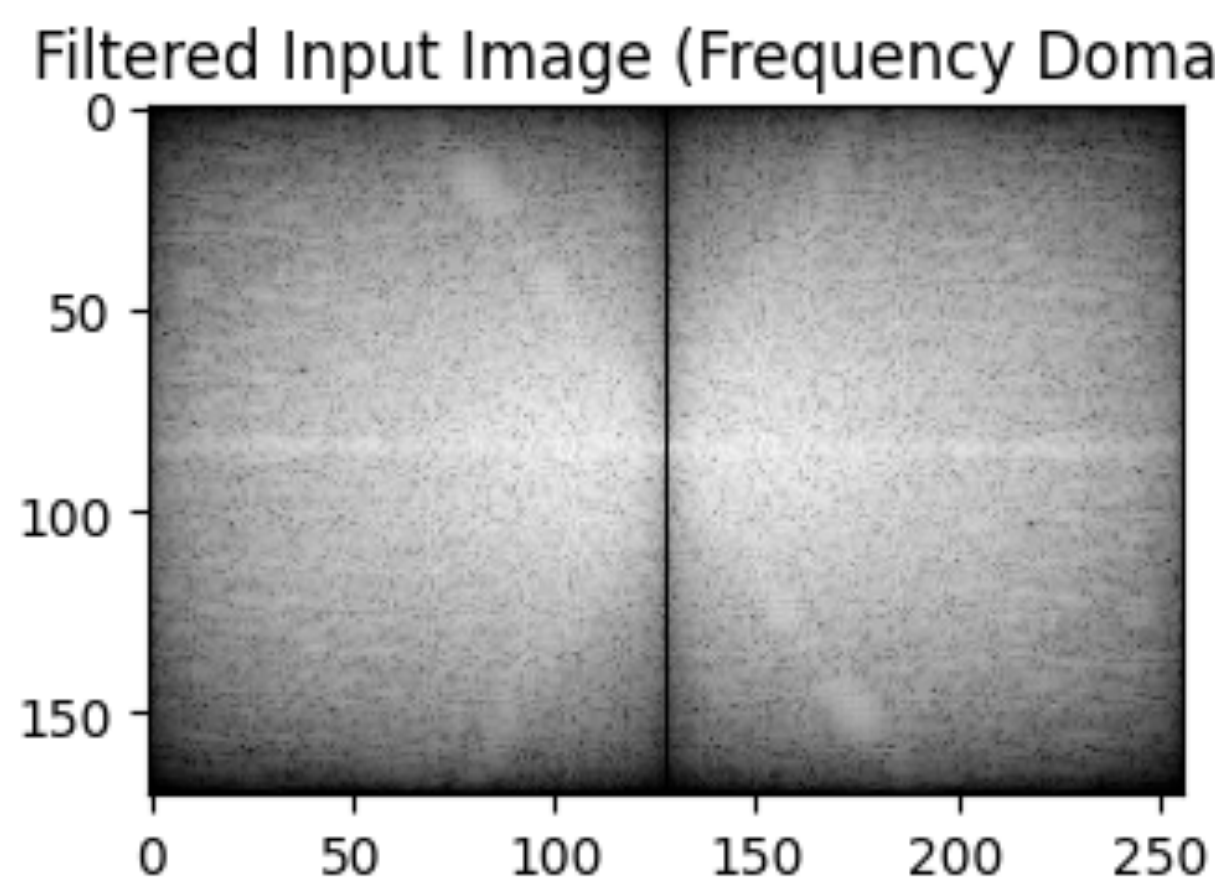
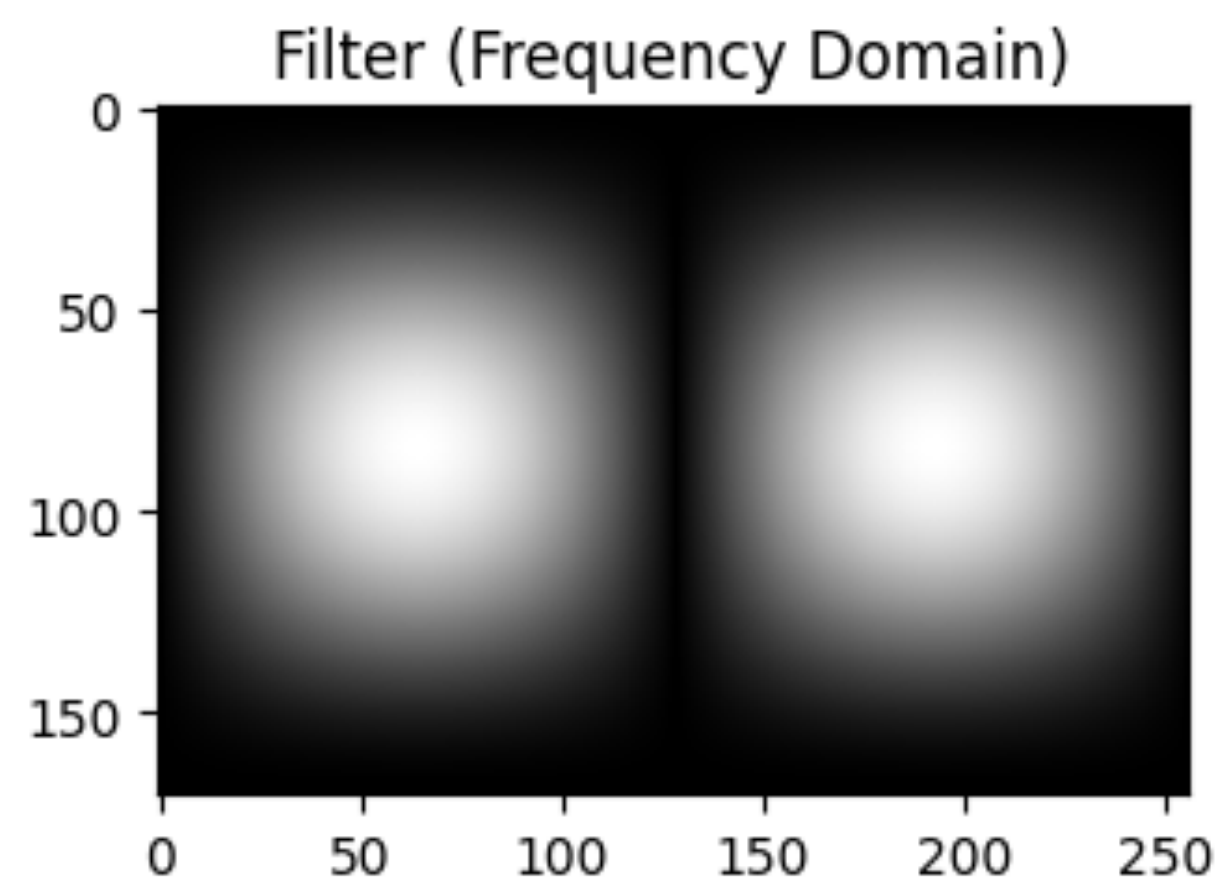
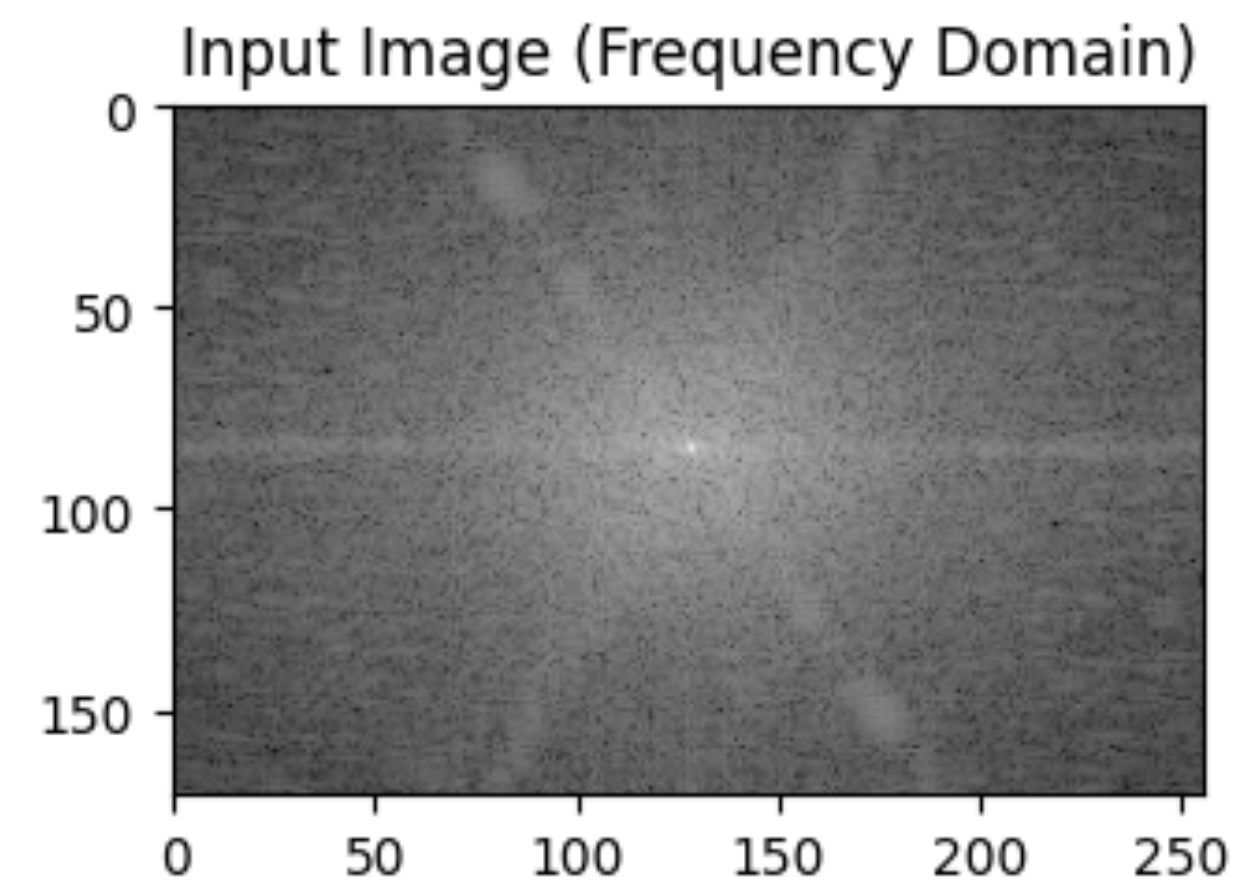
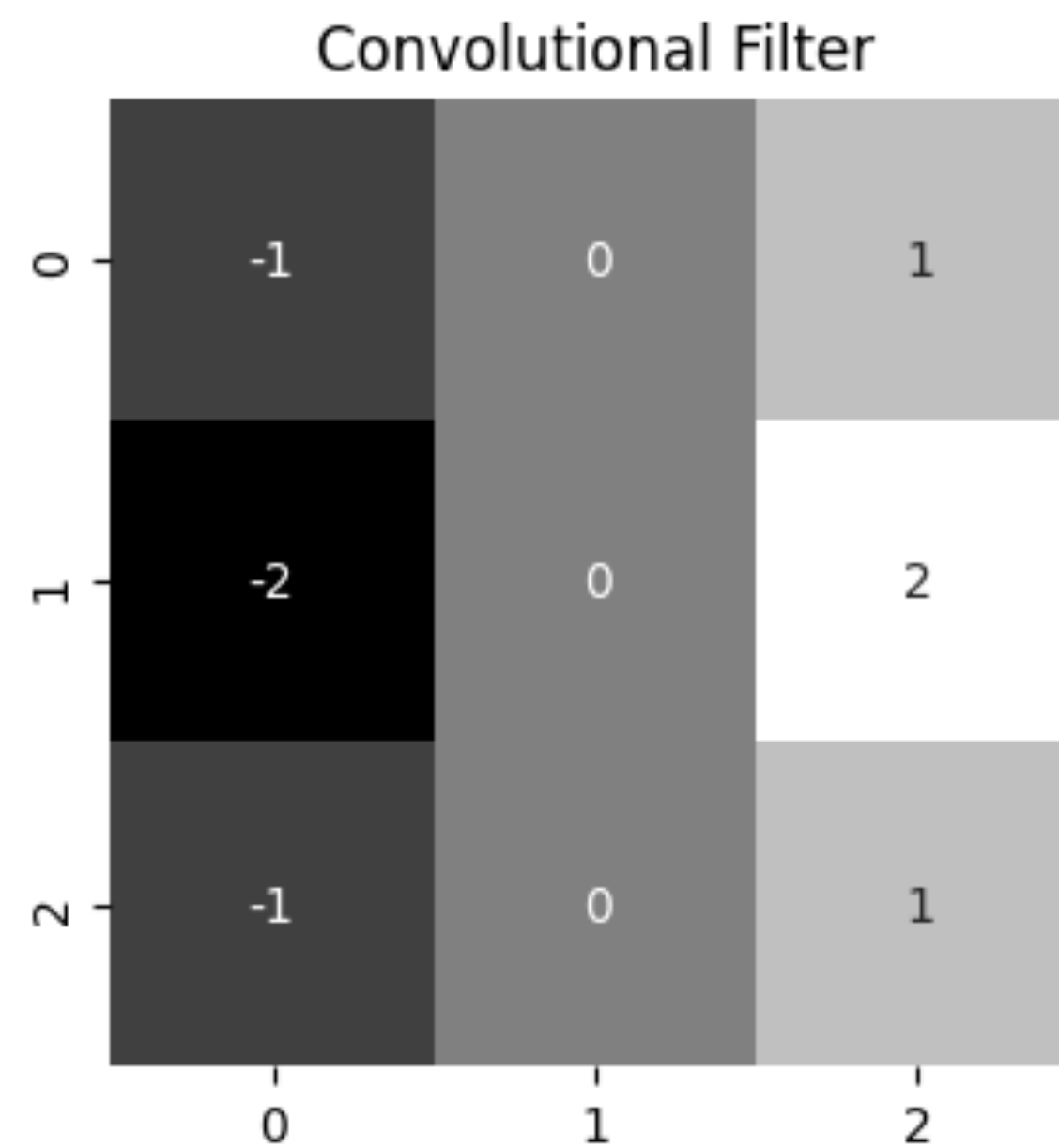
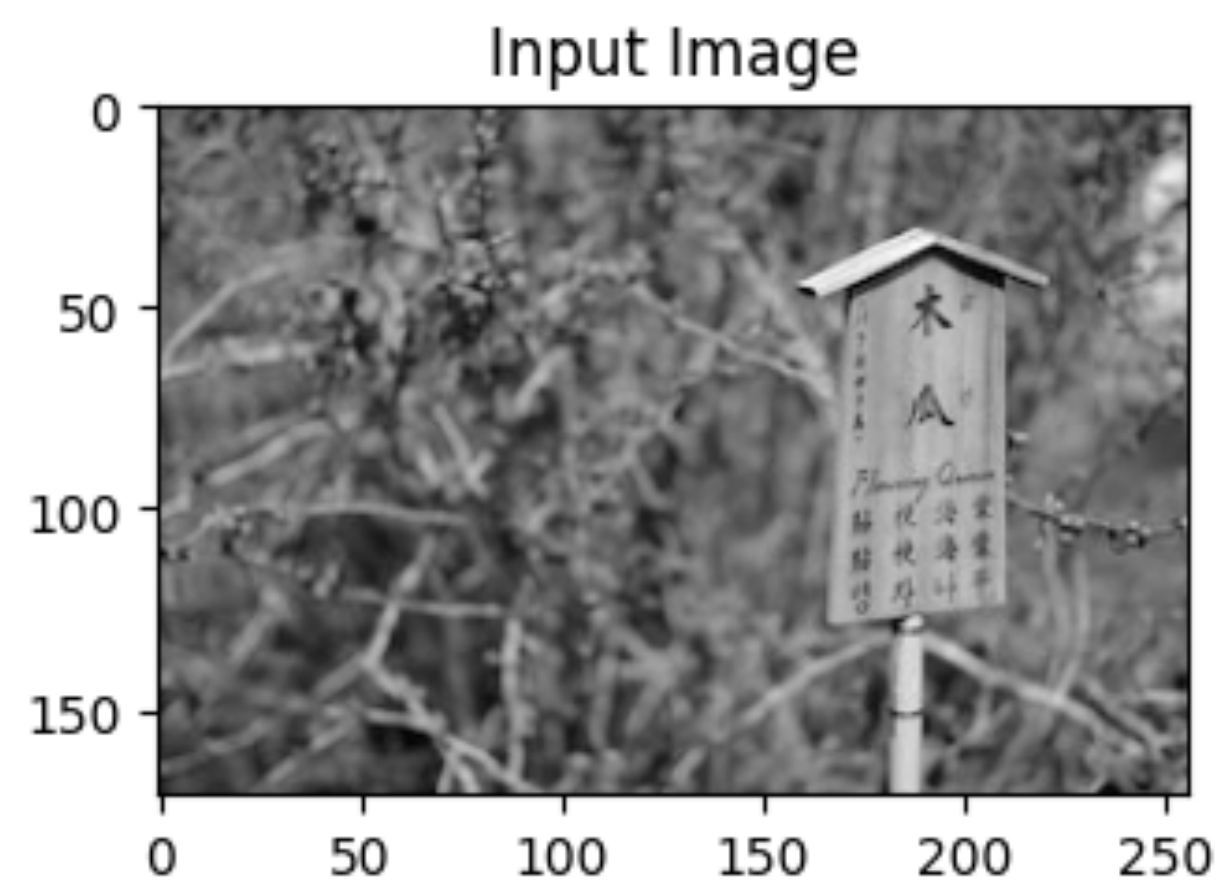




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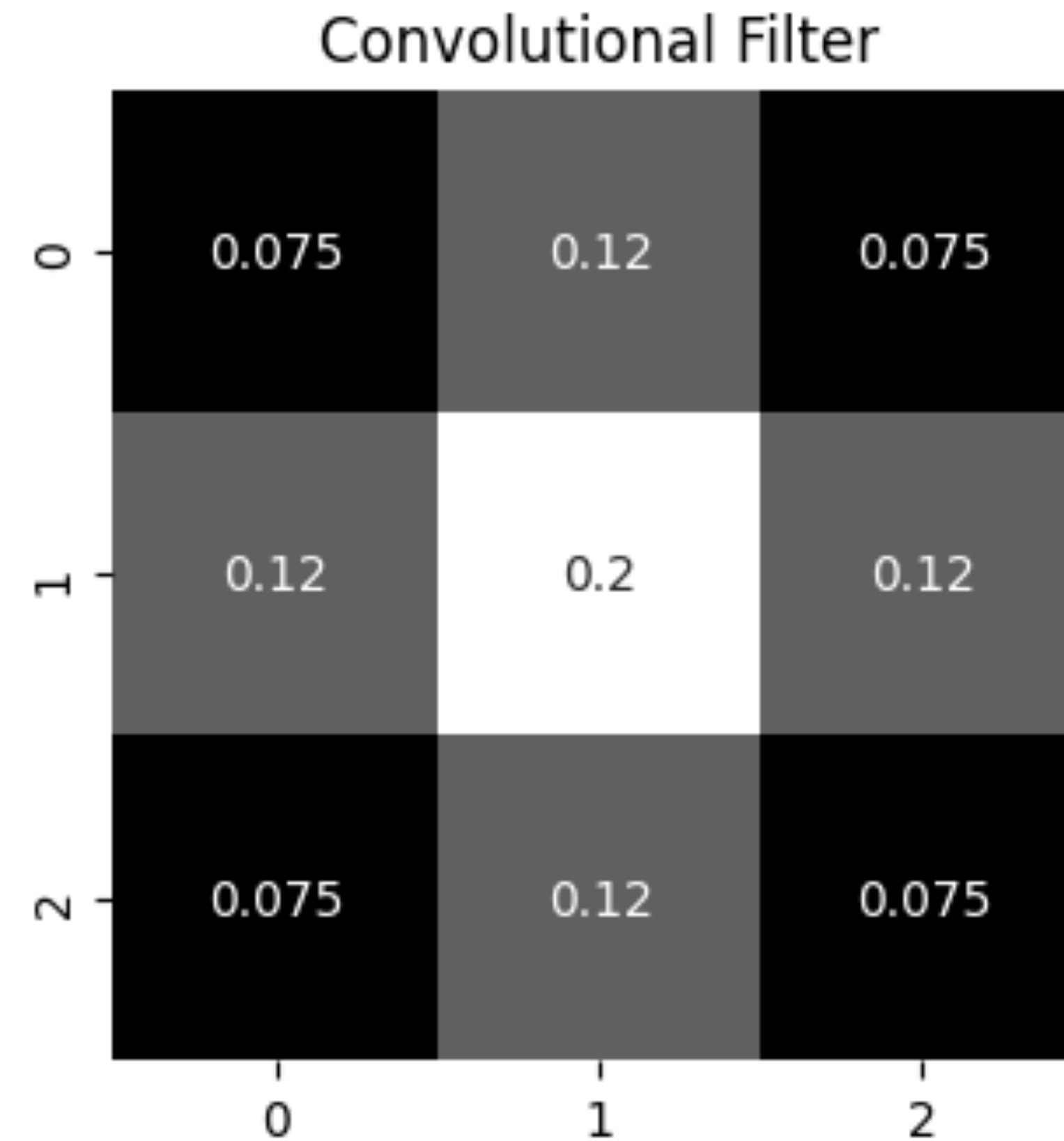
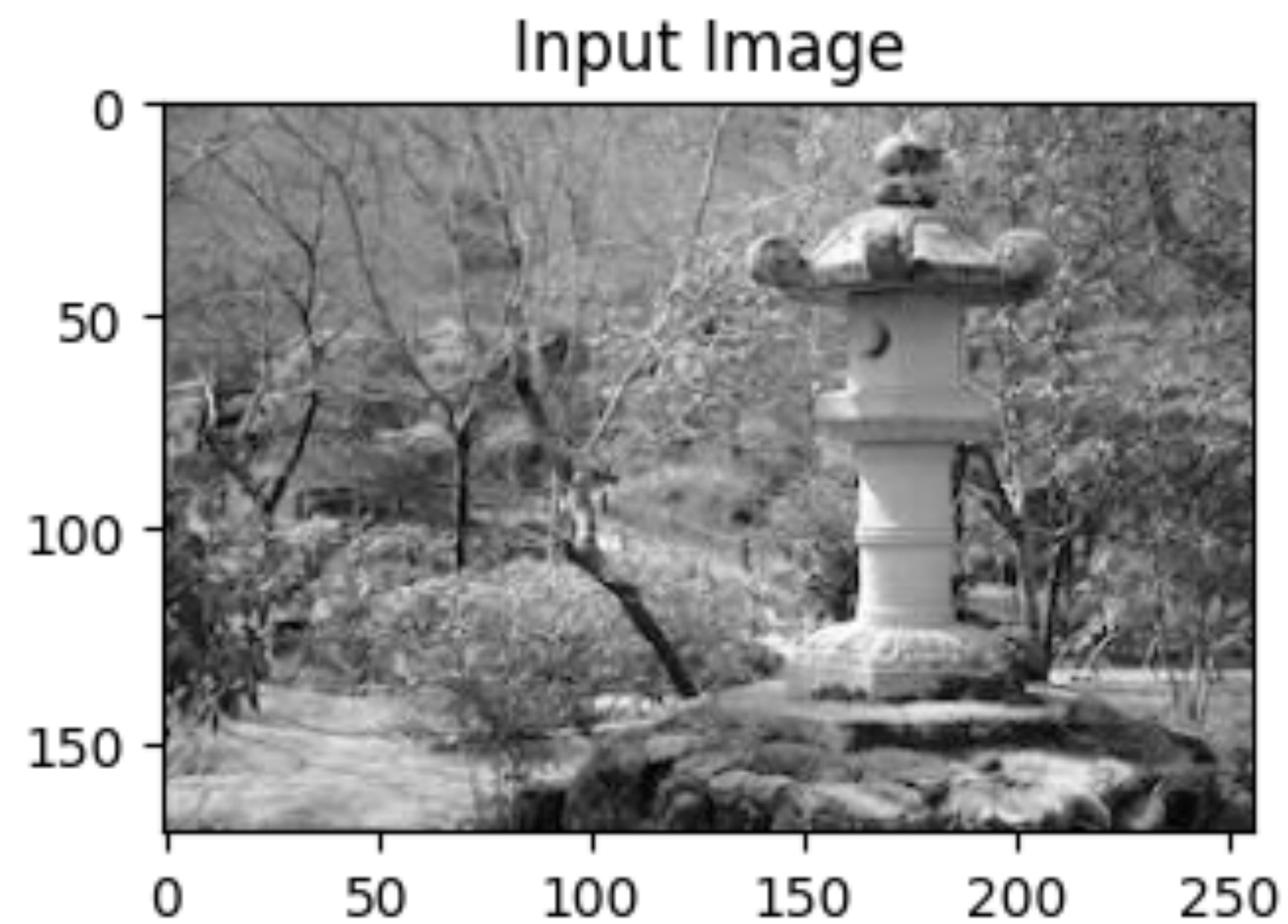
Filtro Sobel

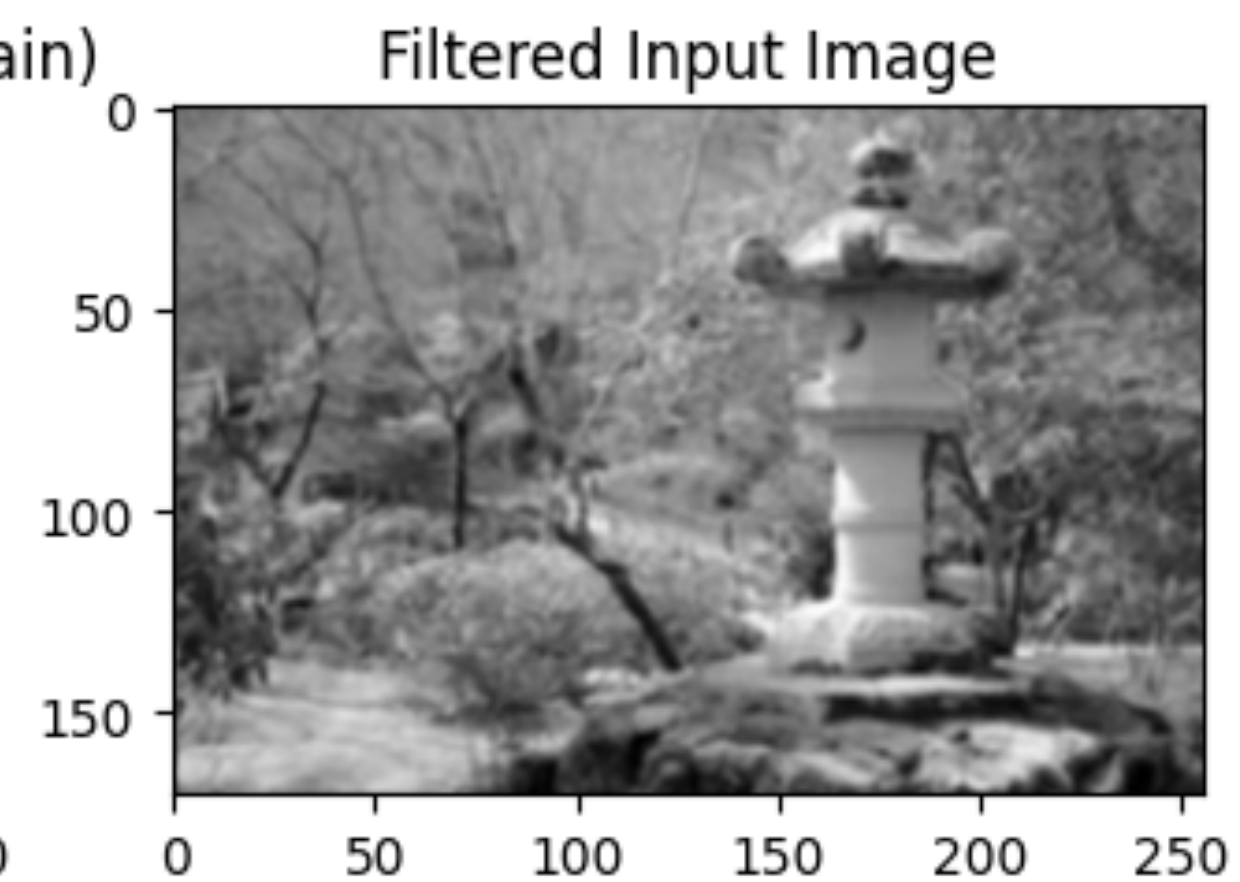
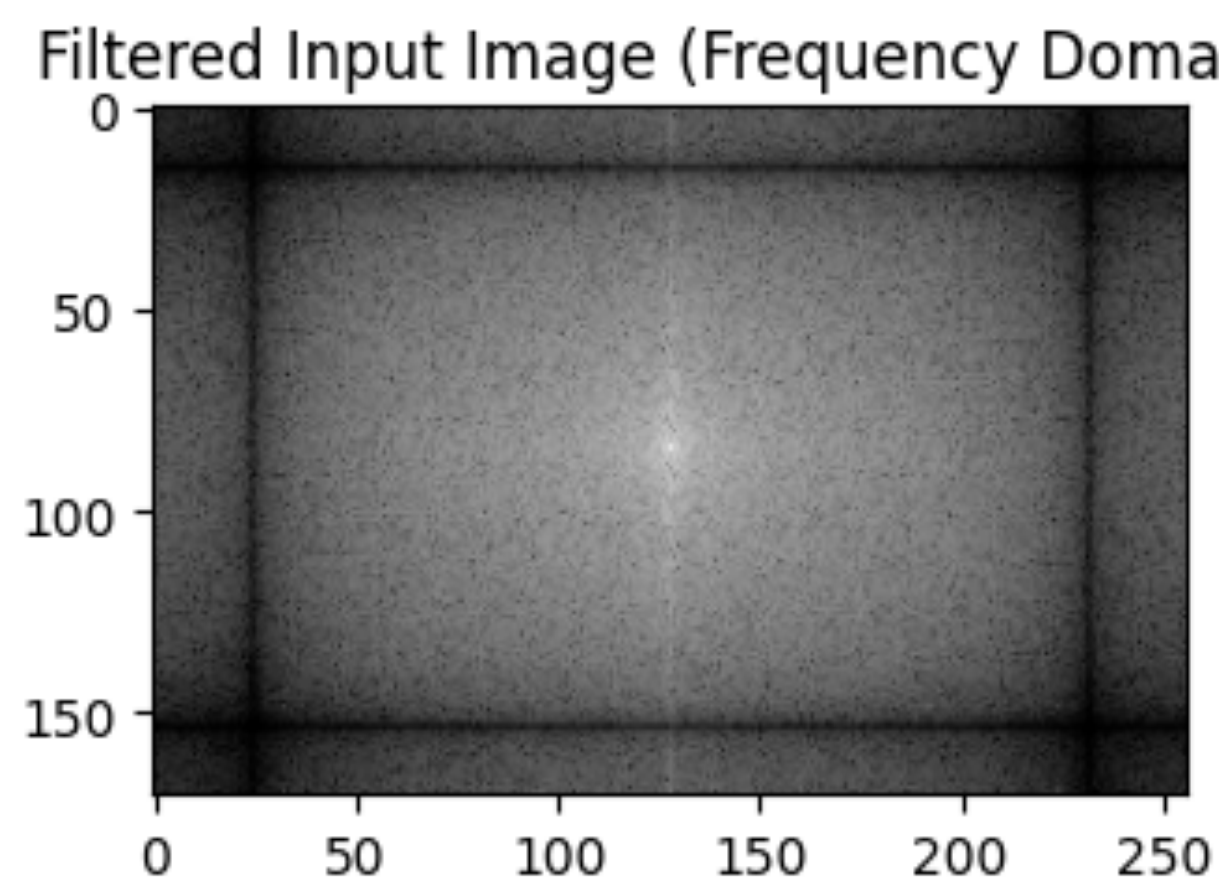
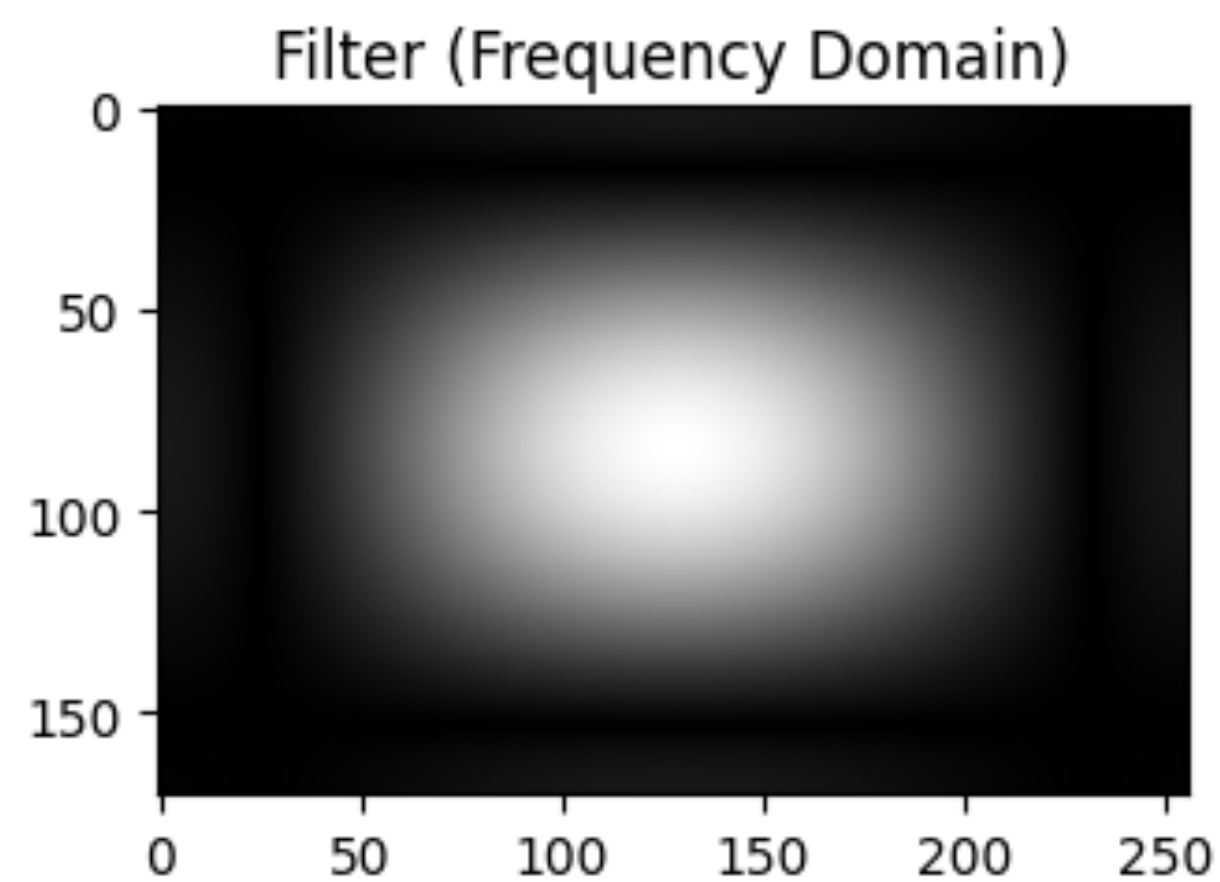
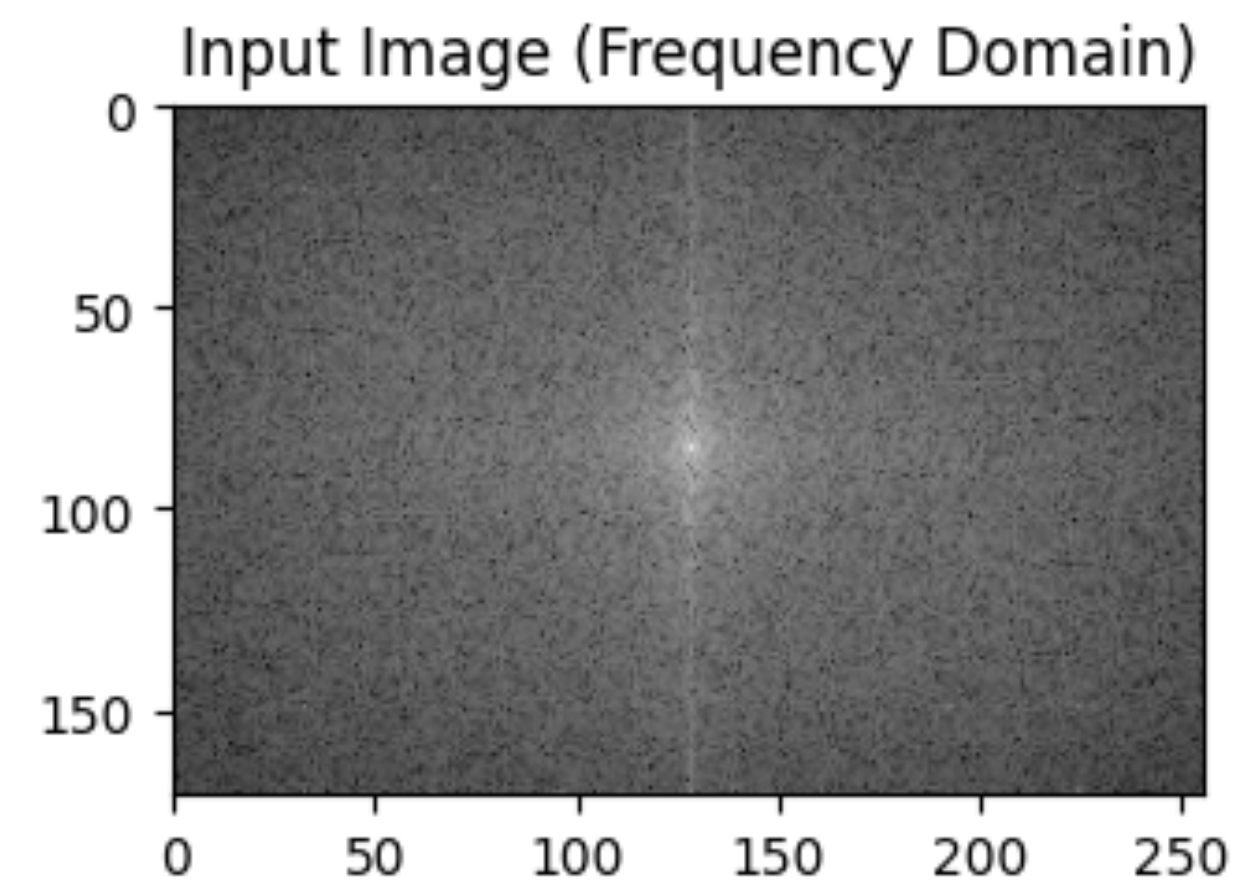
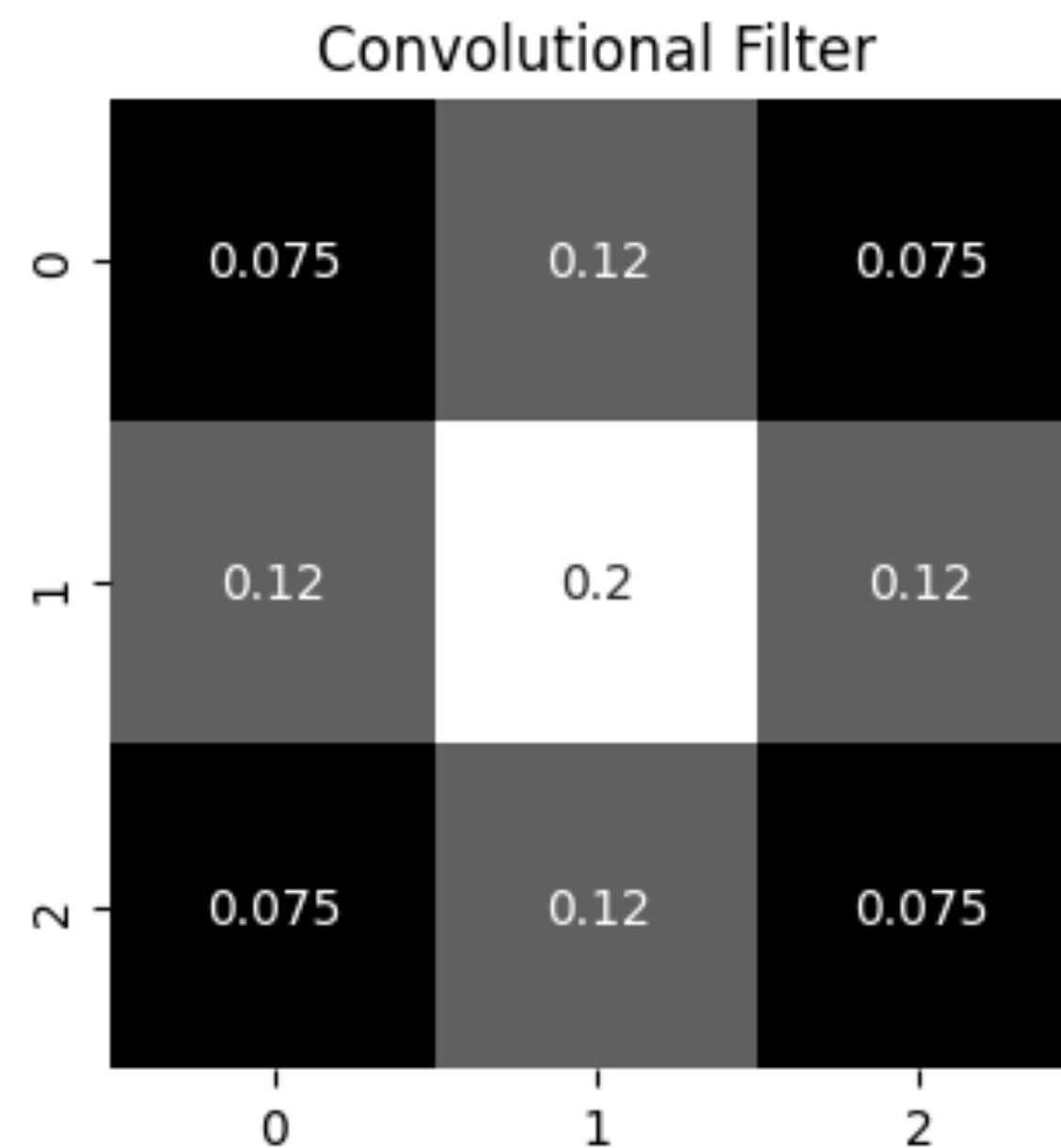
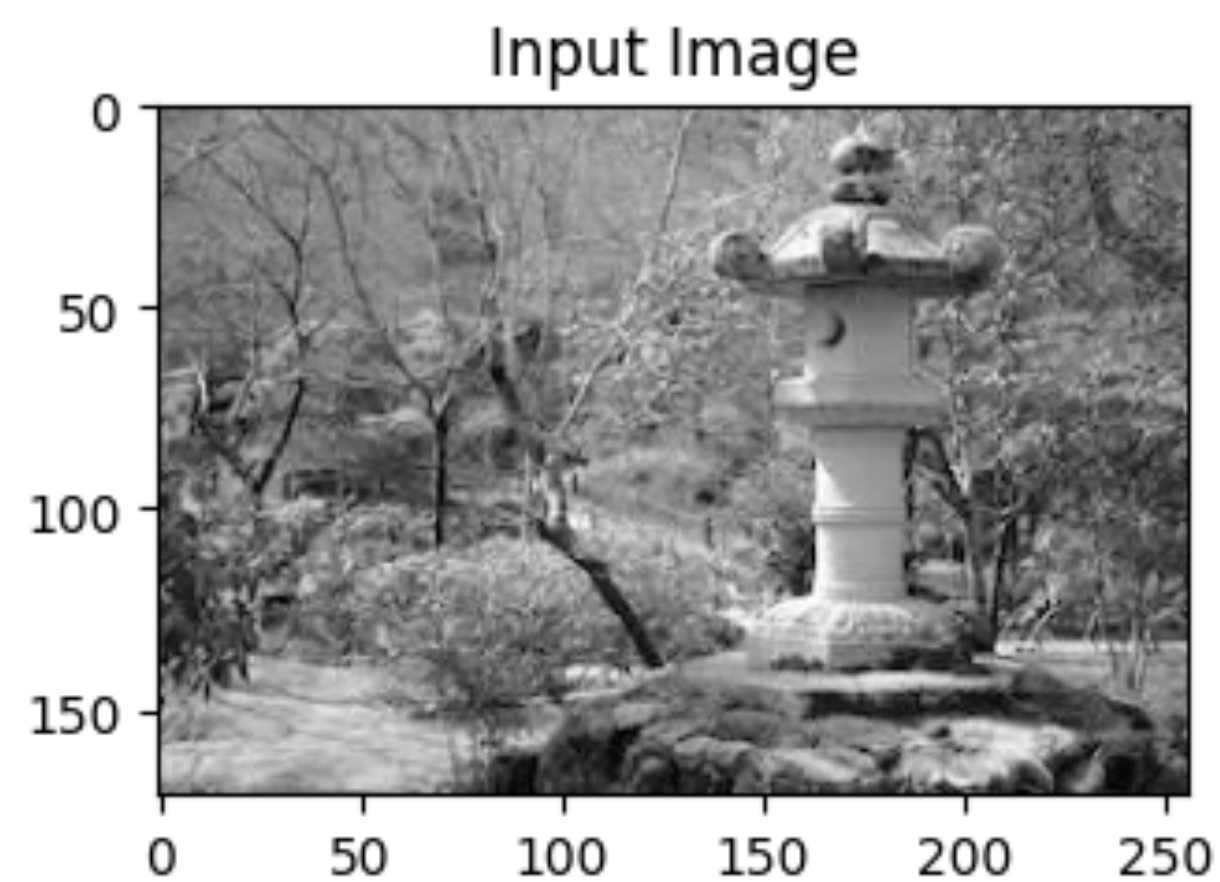




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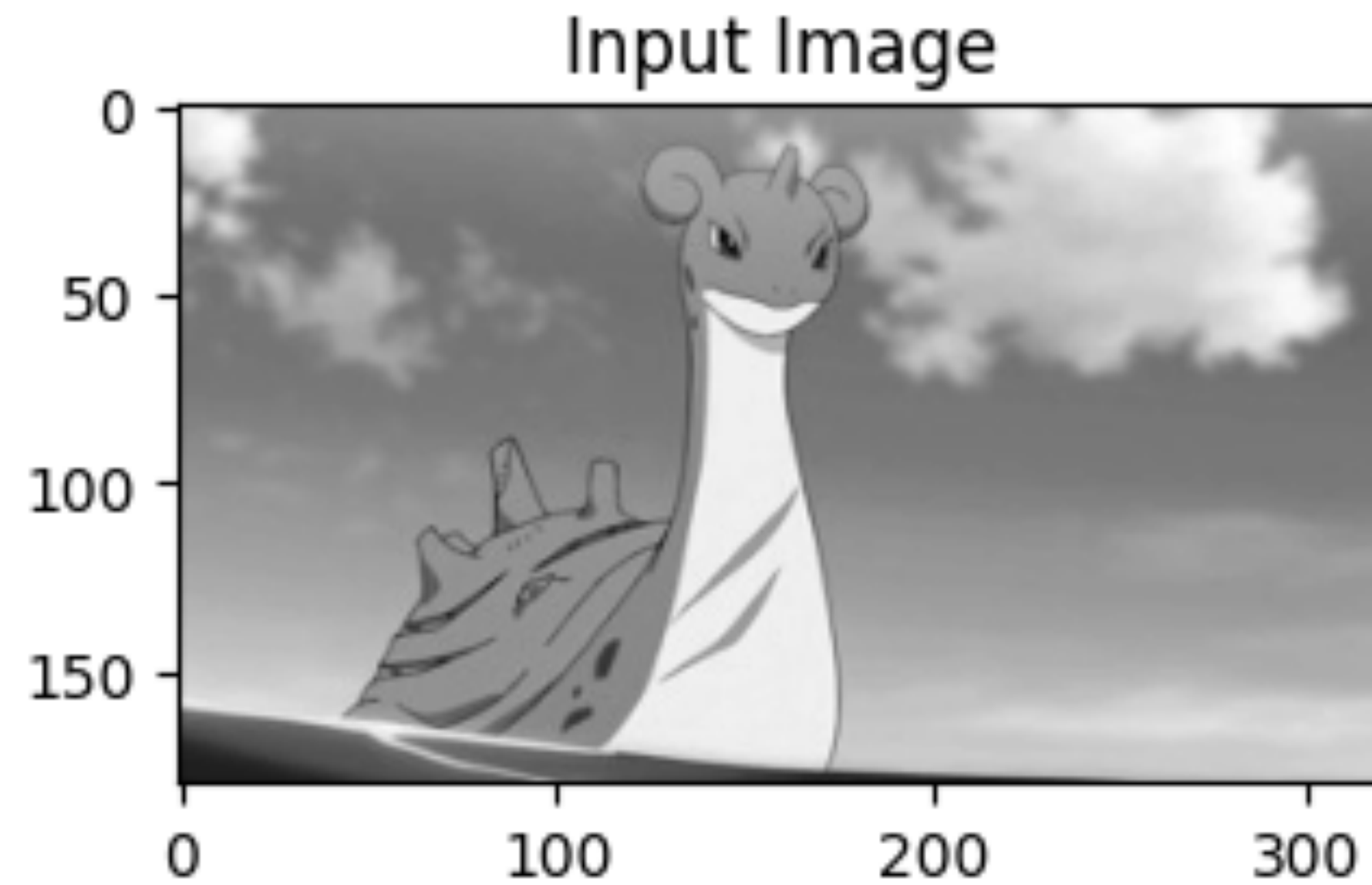
Filtro Gaussiano





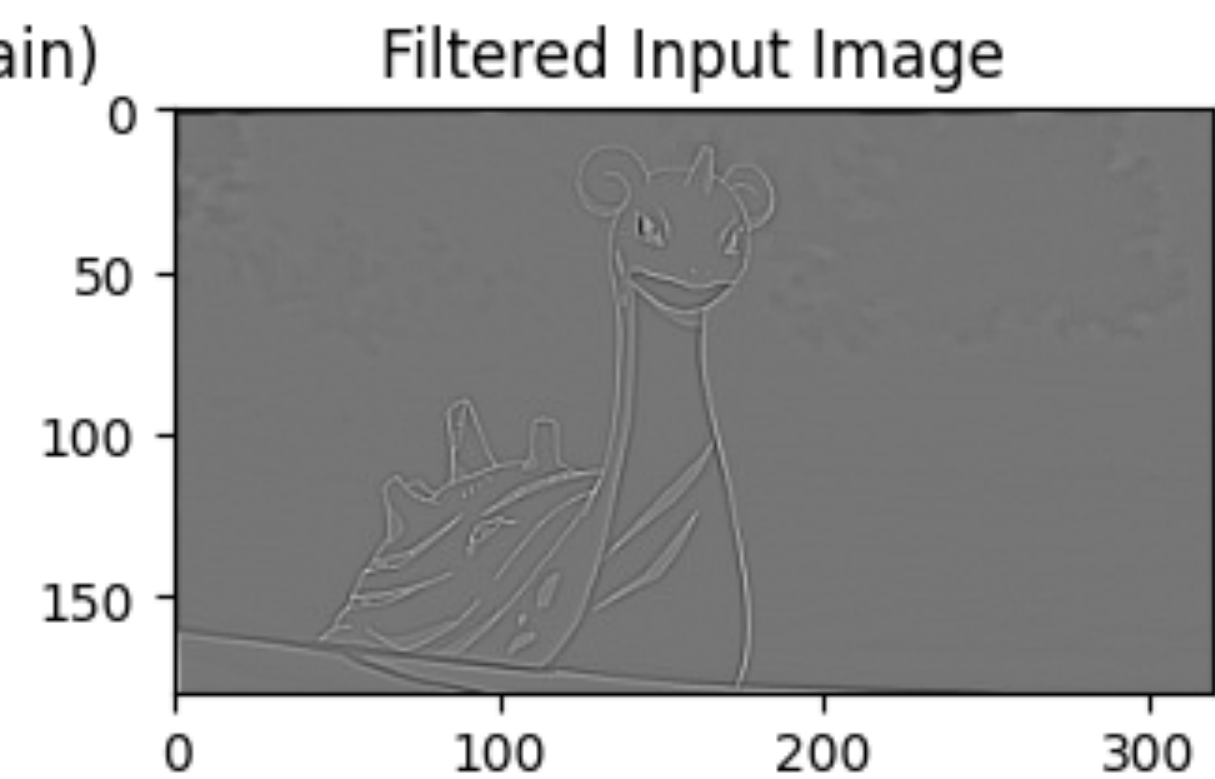
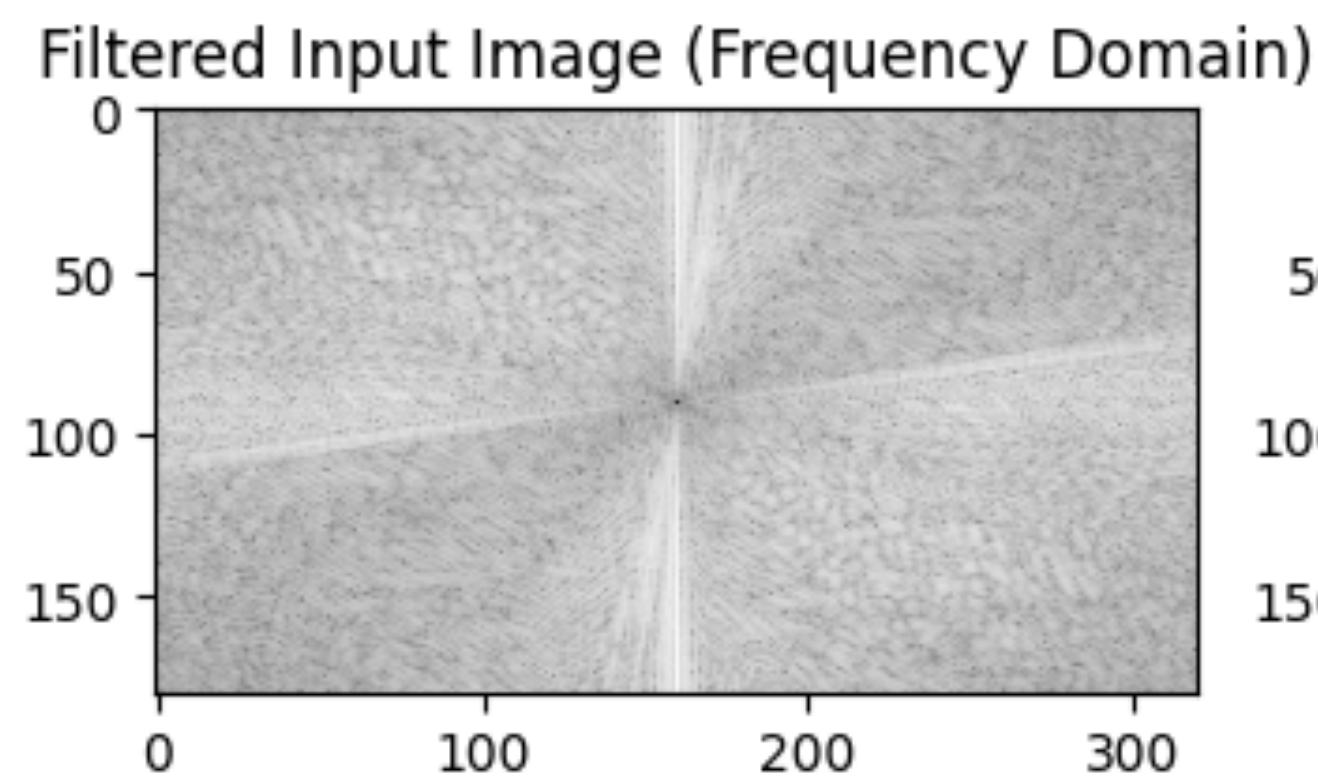
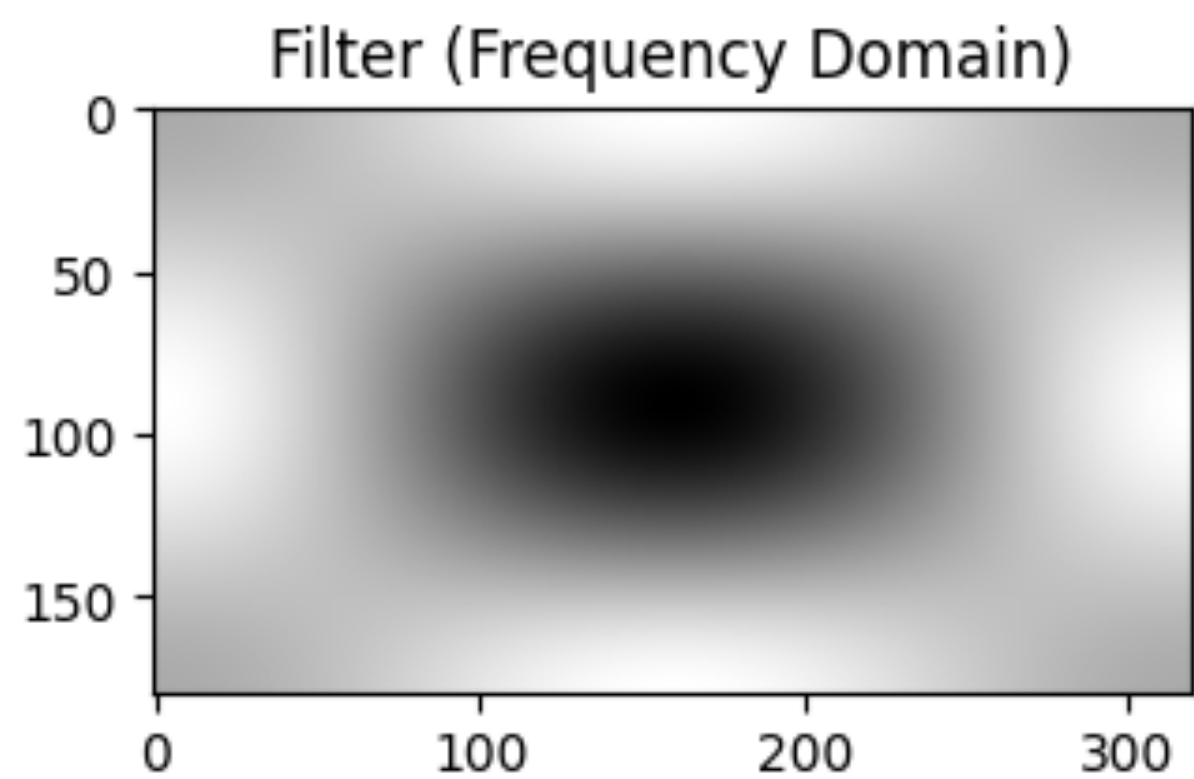
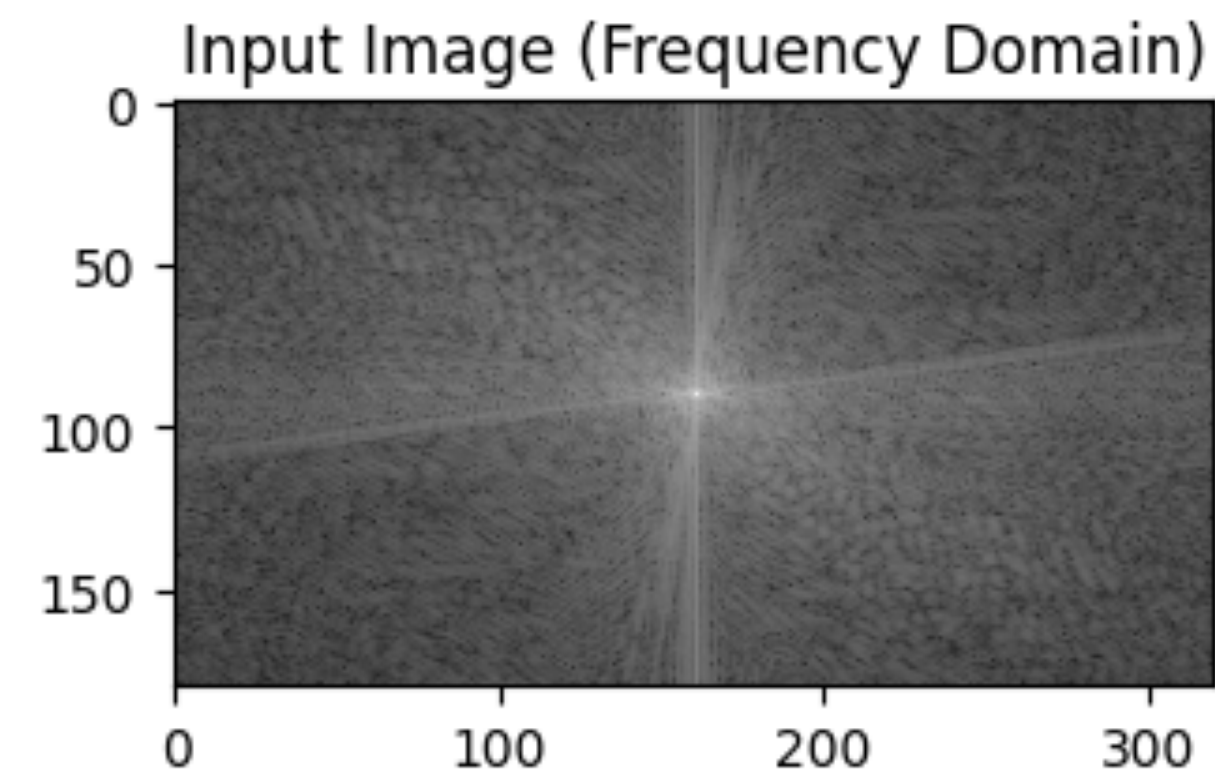
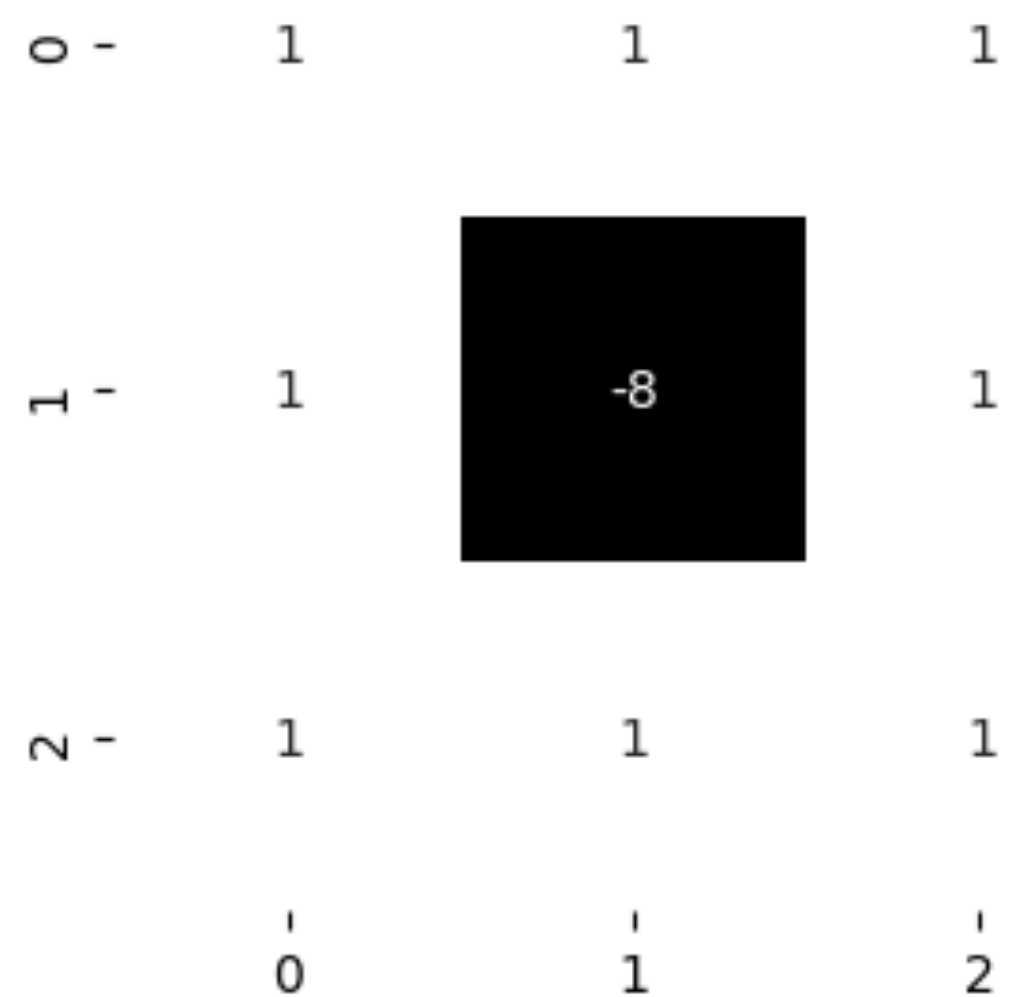
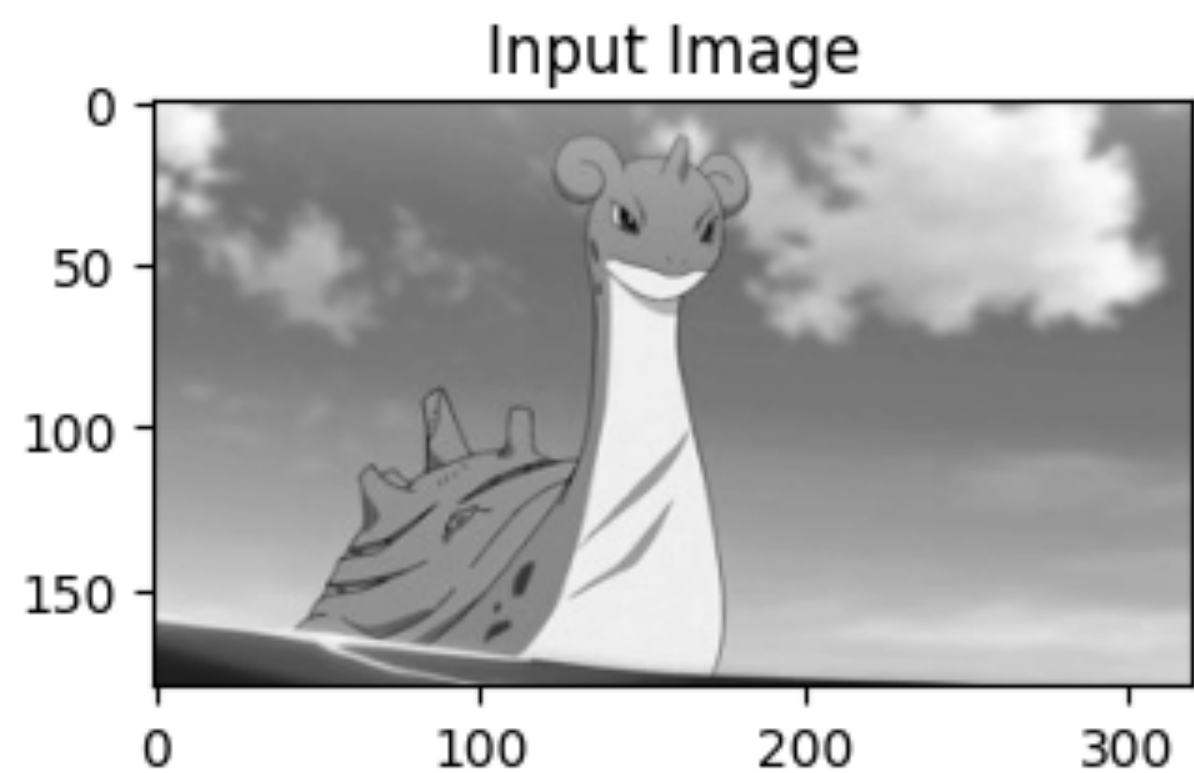
Filtros no Domínio das Frequências

Filtro de Laplace



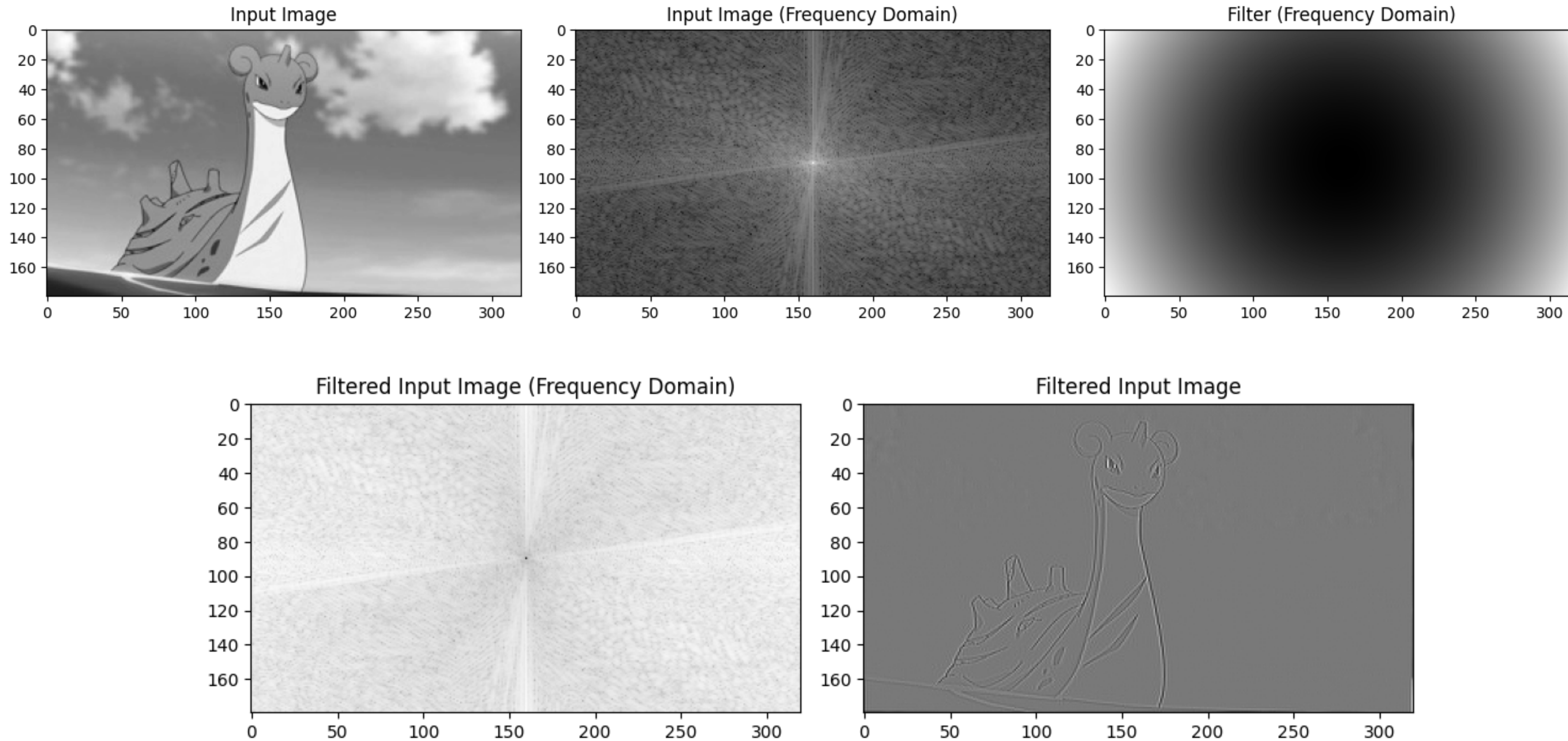
0 -	1	1	1
1 -	1	-8	1
2 -	1	1	1
	0	1	2

Convolutional Filter



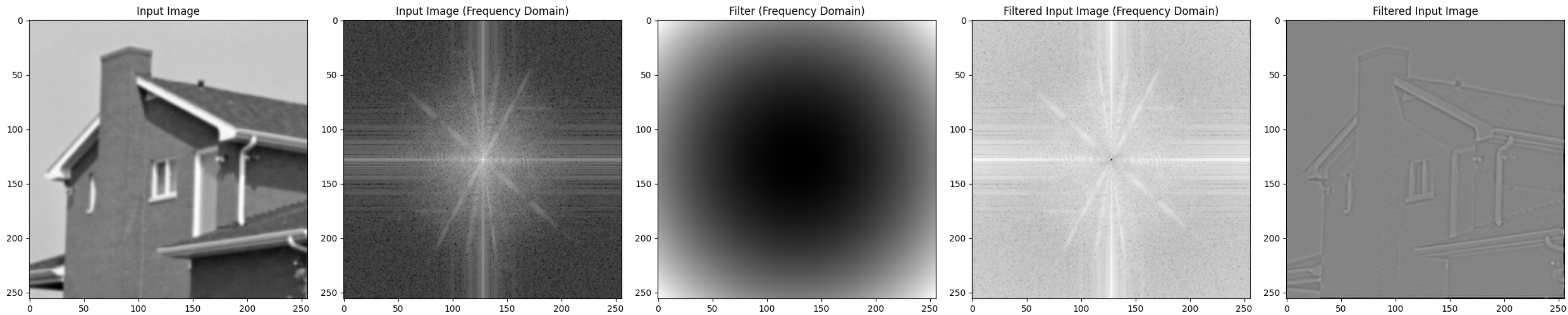
Filtros (perfeitos) no Domínio das Frequências

Filtro de Laplace



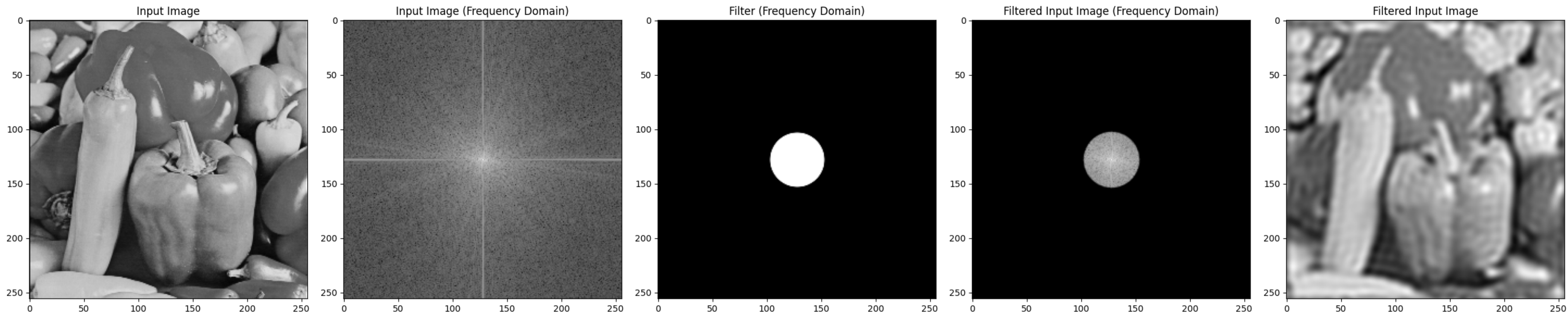
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Filtro de Laplace



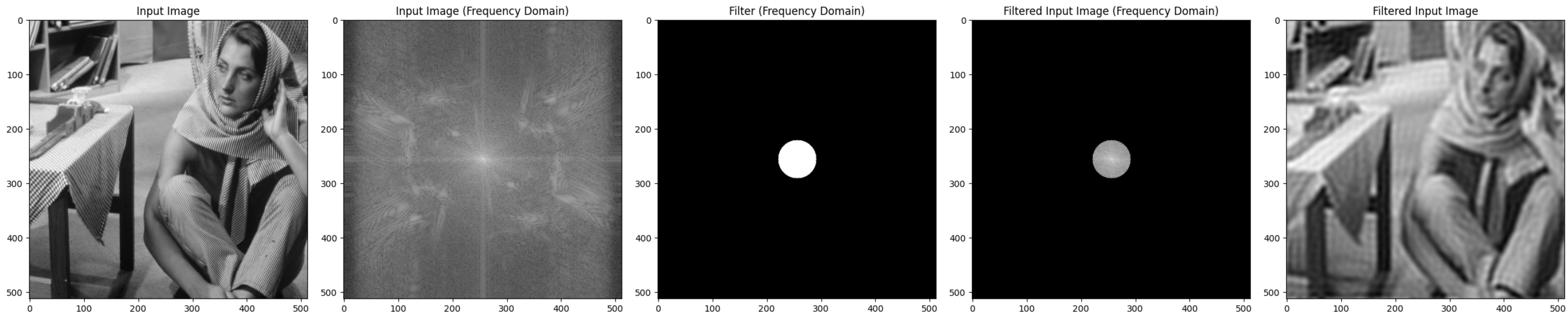
Filtros (perfeitos) no Domínio das Frequências

Filtro passa-baixa perfeito



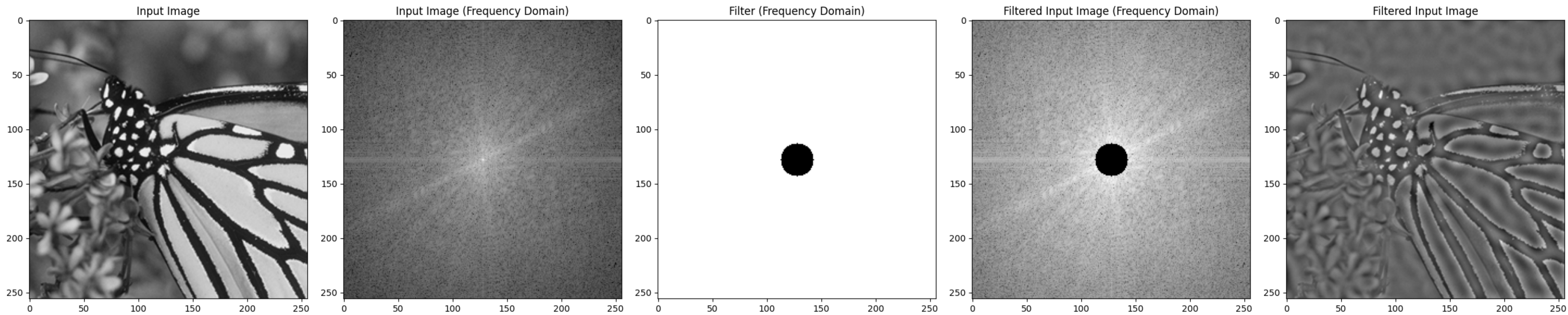
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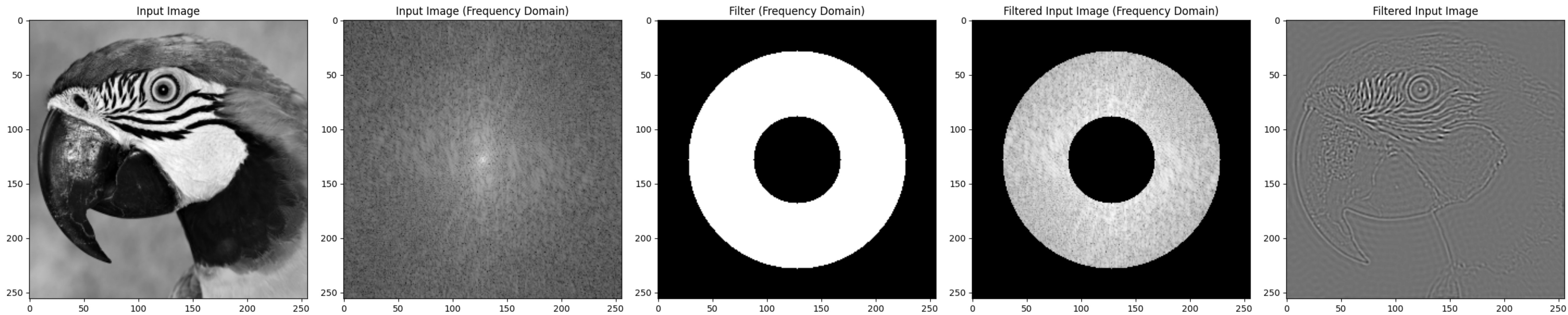
Filtros (perfeitos) no Domínio das Frequências

Filtro passa-alta perfeito



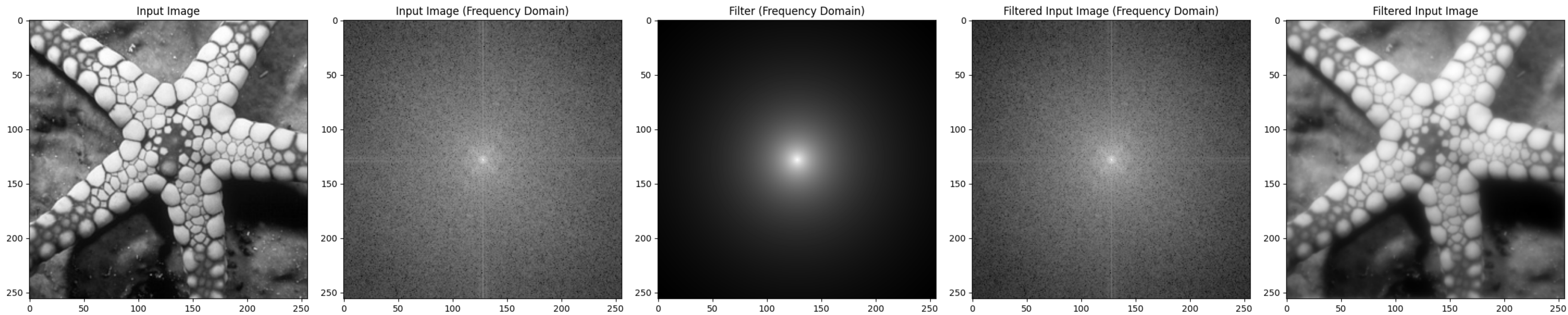
Filtros (perfeitos) no Domínio das Frequências

Filtro passa-faixa perfeito



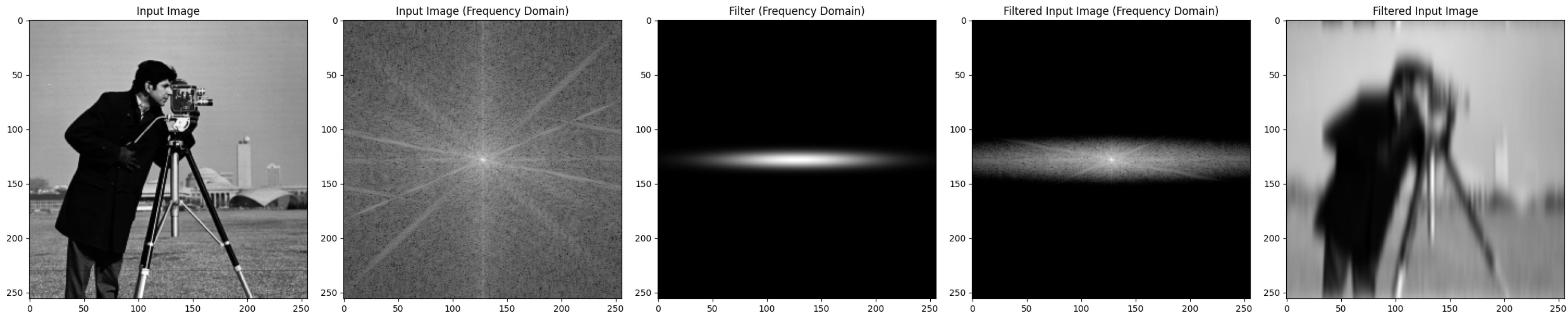
Filtros (perfeitos) no Domínio das Frequências

Filtro gaussiano



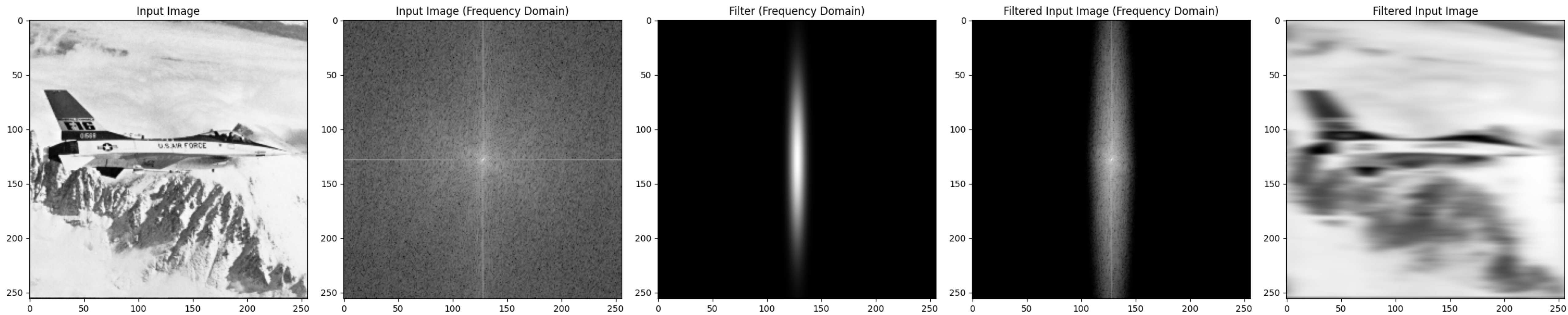
Filtros (perfeitos) no Domínio das Frequências

Filtro gaussiano



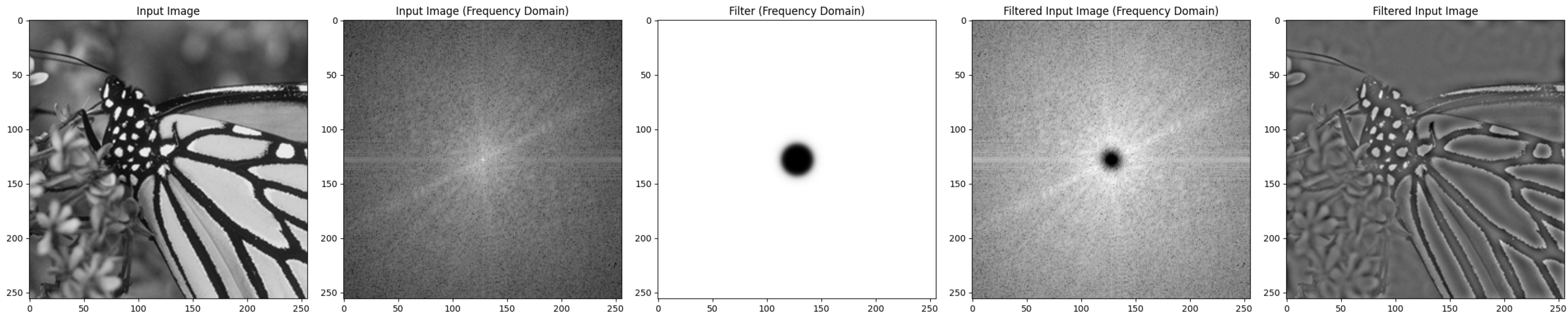
Filtros (perfeitos) no Domínio das Frequências

Filtro gaussiano



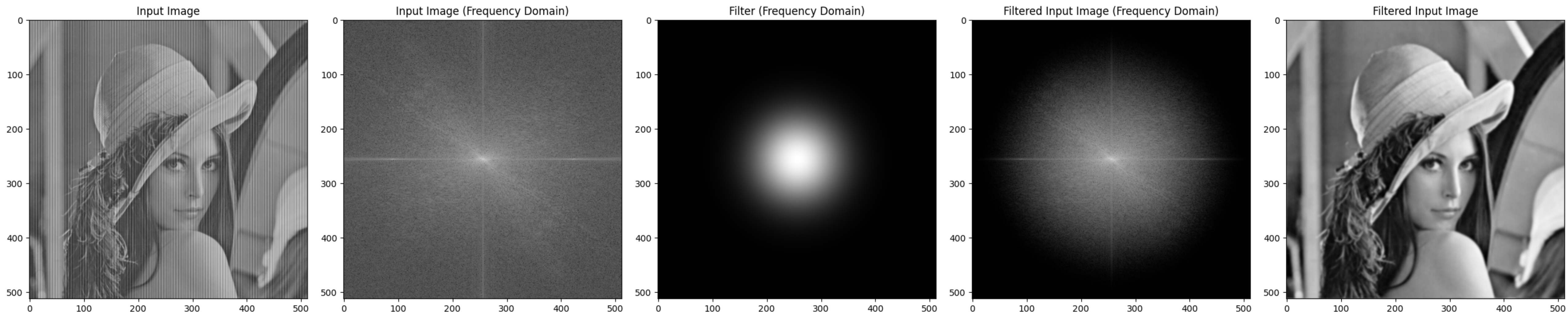
Filtros (perfeitos) no Domínio das Frequências

Filtro de Butterworth passa-alta



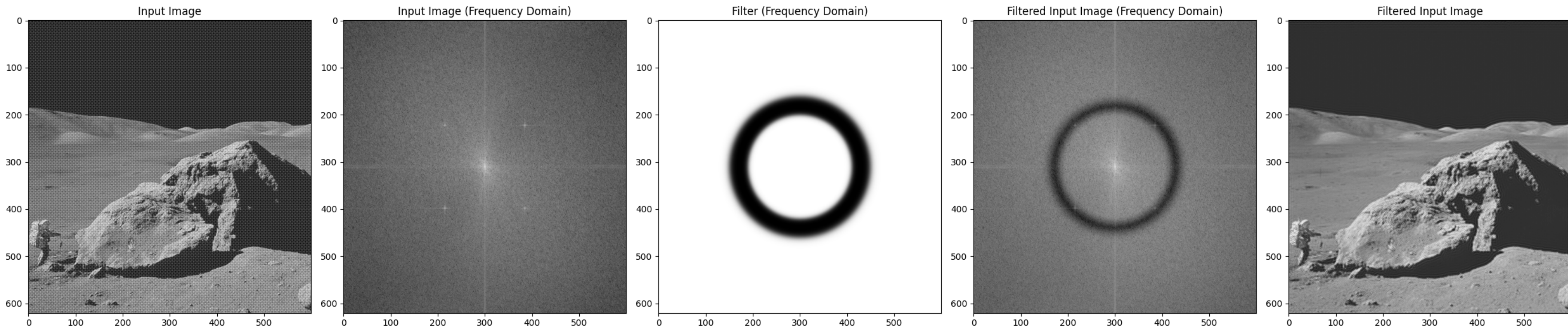
Filtros (perfeitos) no Domínio das Frequências

Filtro de Butterworth passa-baixa



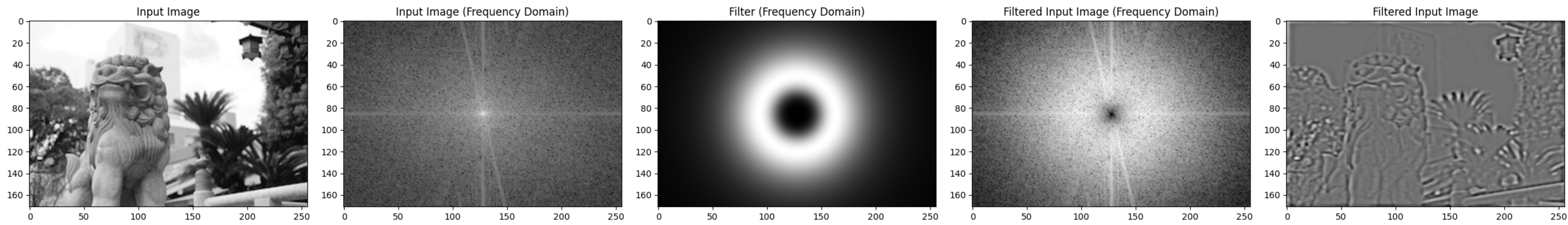
Filtros (perfeitos) no Domínio das Frequências

Filtro de Butterworth rejeita-faixa



Filtros (perfeitos) no Domínio das Frequências

Filtro de Butterworth passa-faixa



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